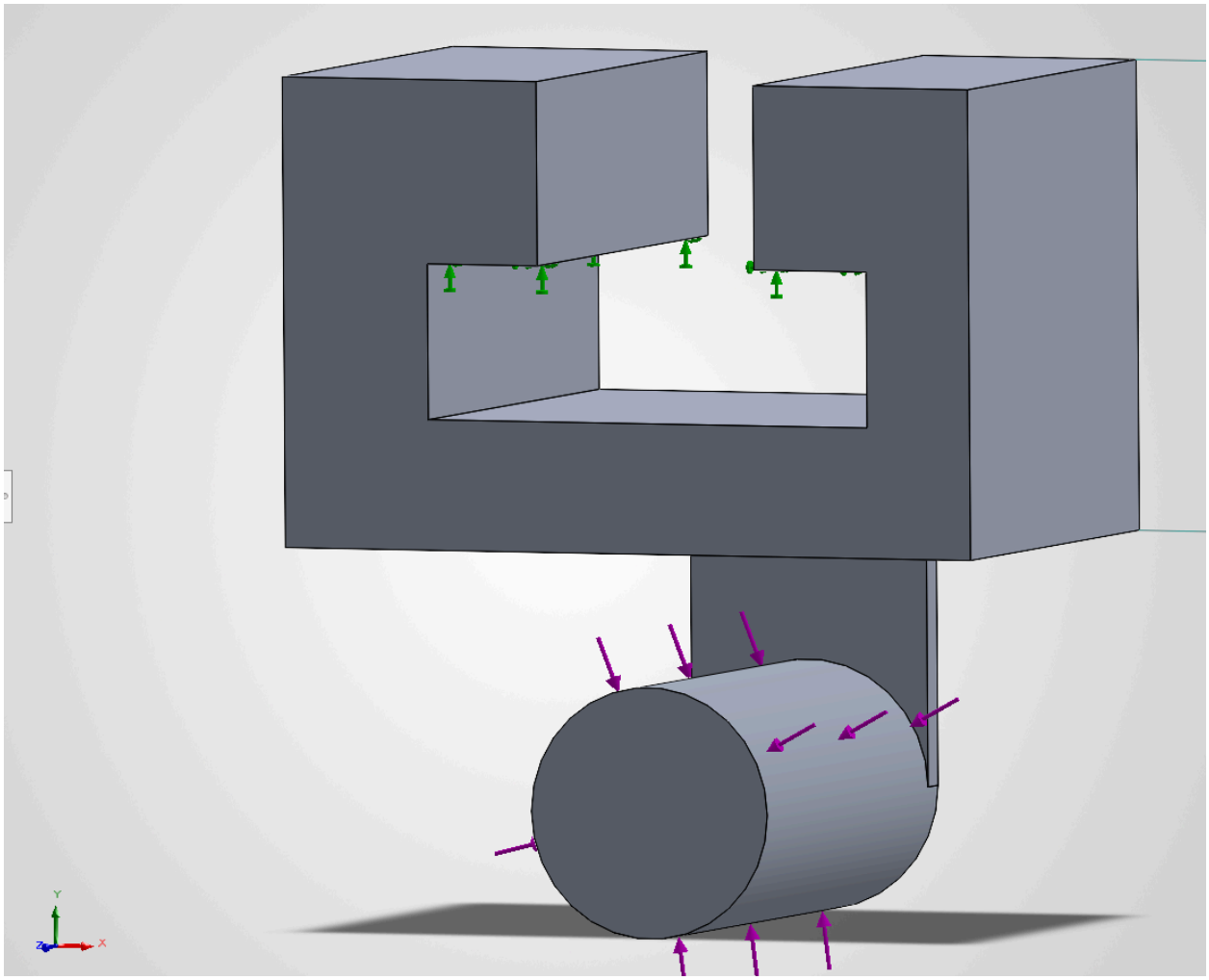
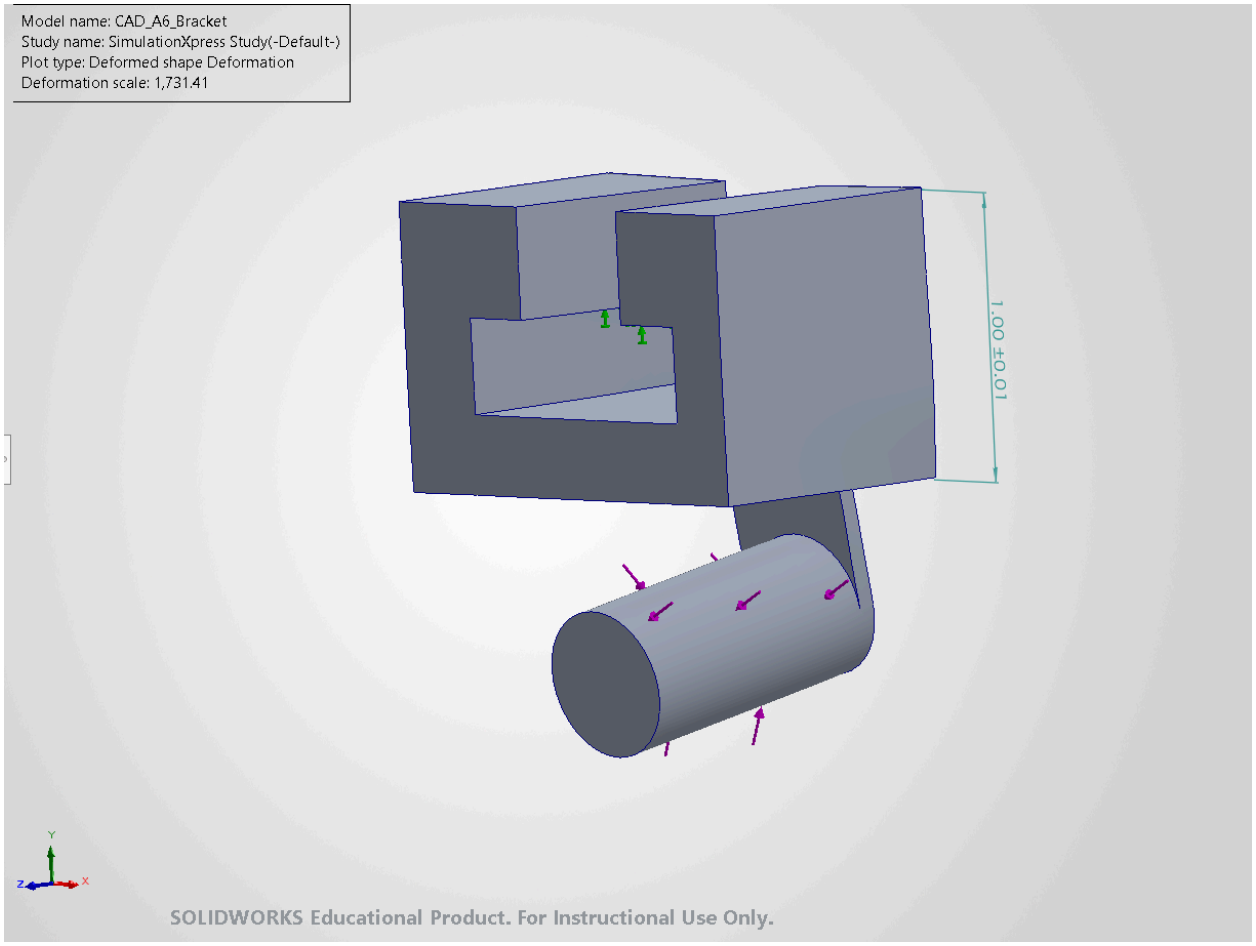
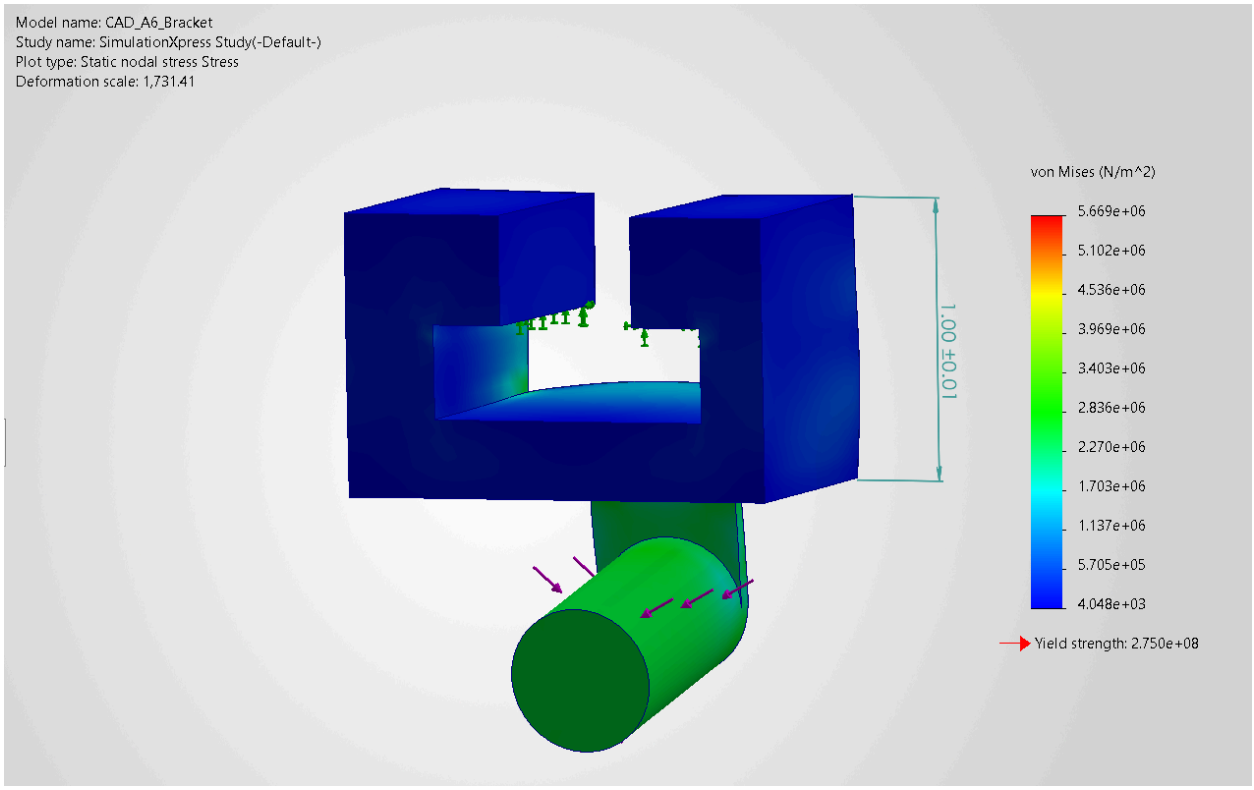
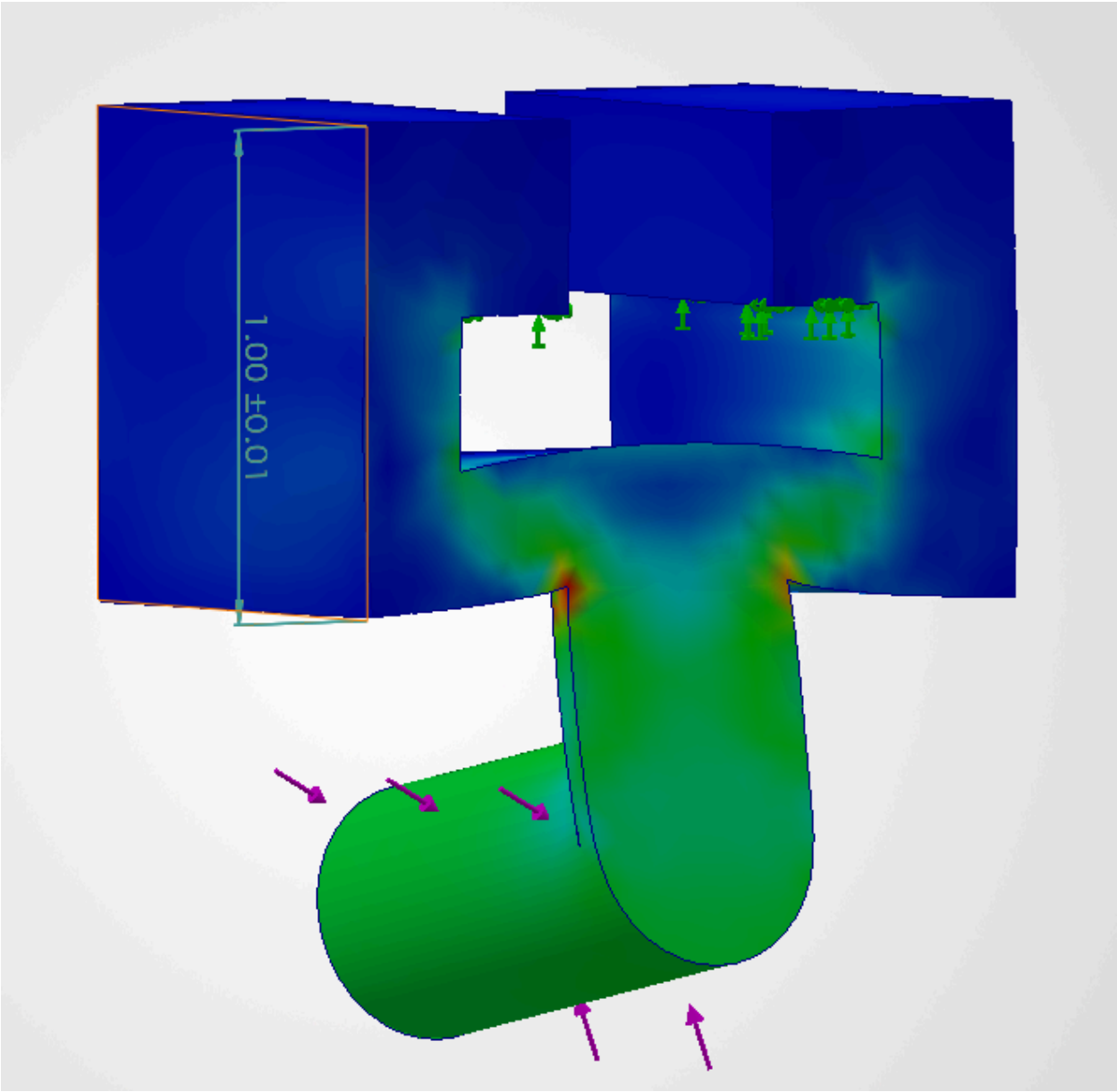


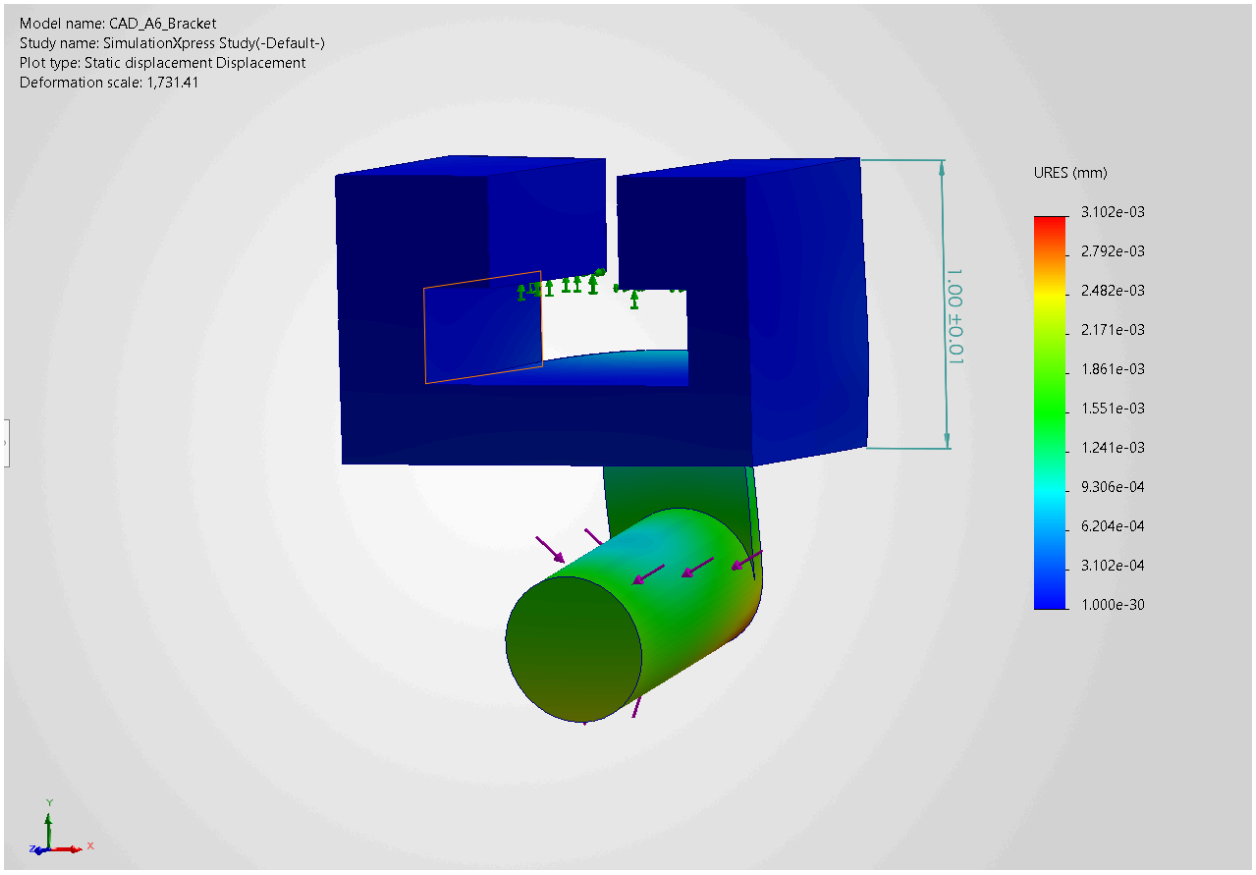


Deformation

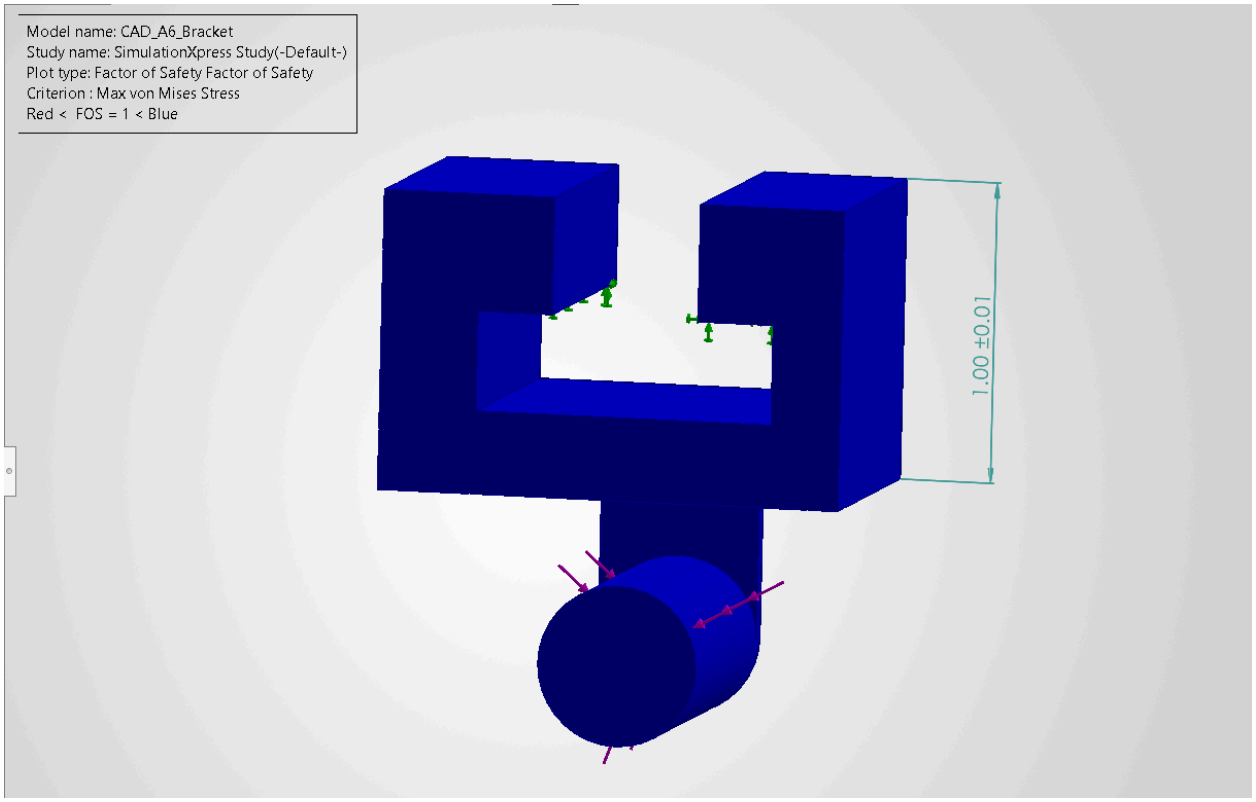

Model name: CAD_A6_Bracket
Study name: SimulationXpress Study(-Default-)
Plot type: Static nodal stress Stress
Deformation scale: 1,731.41
1.00 +0.01
von Mises (N/m^2)
5.669e+06
5.102e+06
4.536e+06
3.969e+06
3.403e+06
2.836e+06
2.270e+06
1.703e+06
1.137e+06
5.705e+05
4.048e+03
Yield strength: 2.750e+08



Von-mises stress



Displacement



Factor of Safety

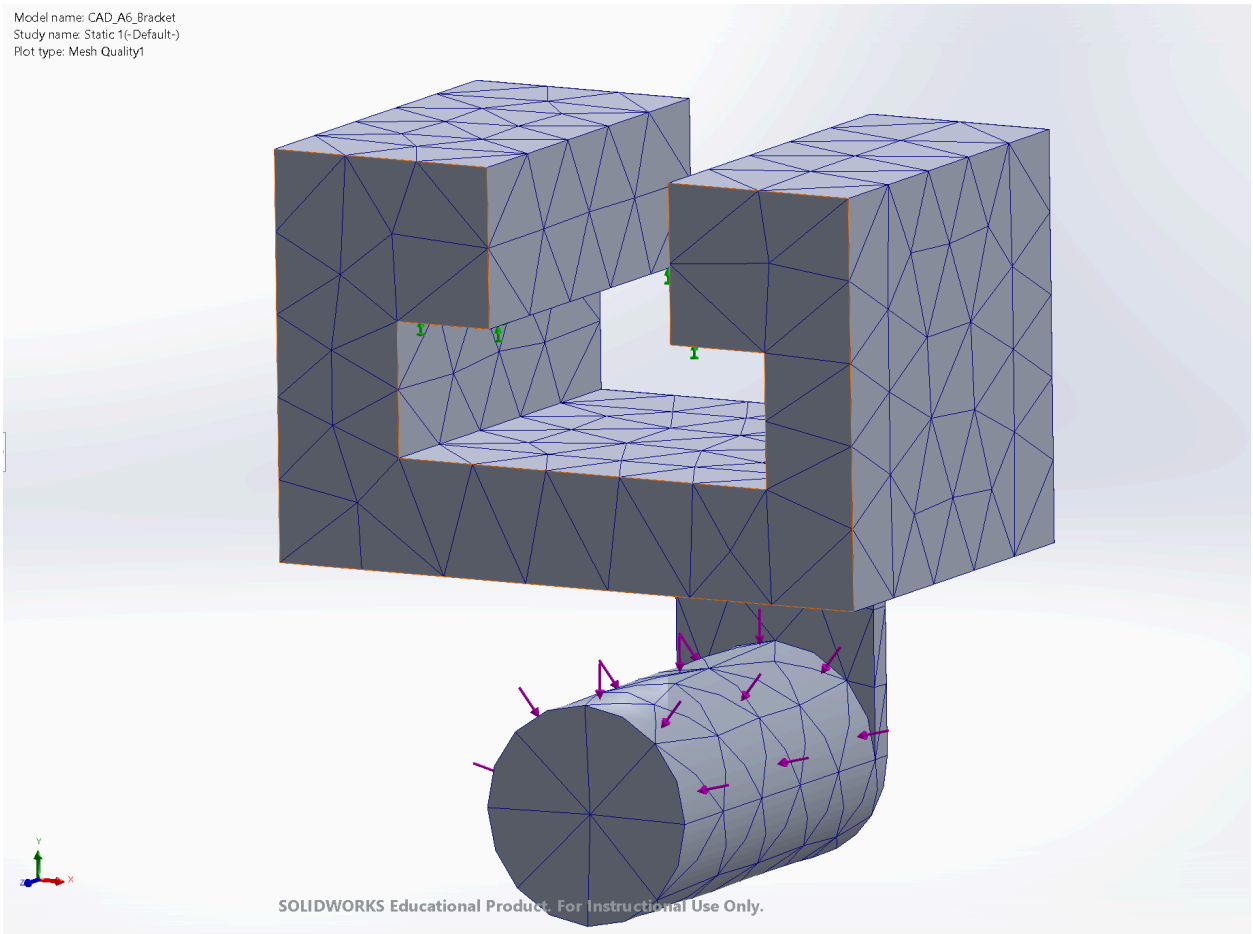
Based on the specified parameters, the lowest factor of safety(FOS) found in your design is 48.5116

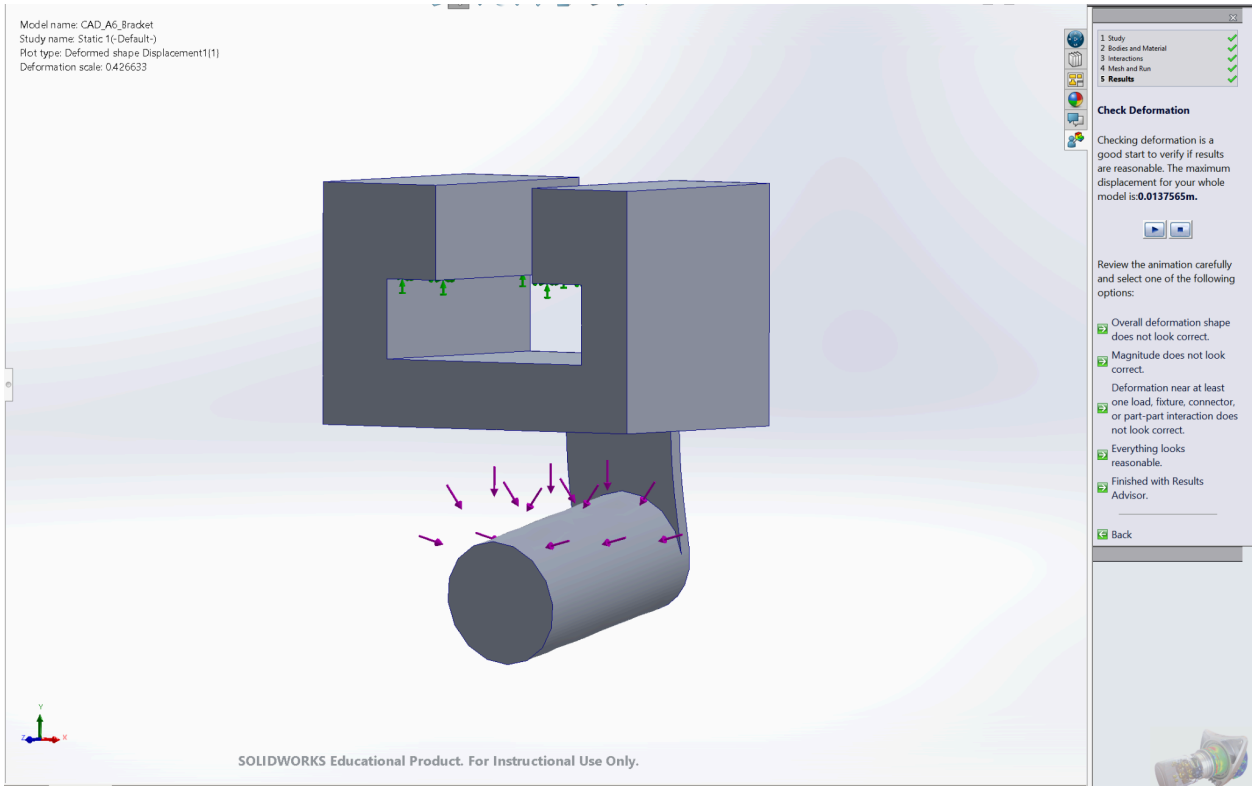
Did not converge.

3D Design

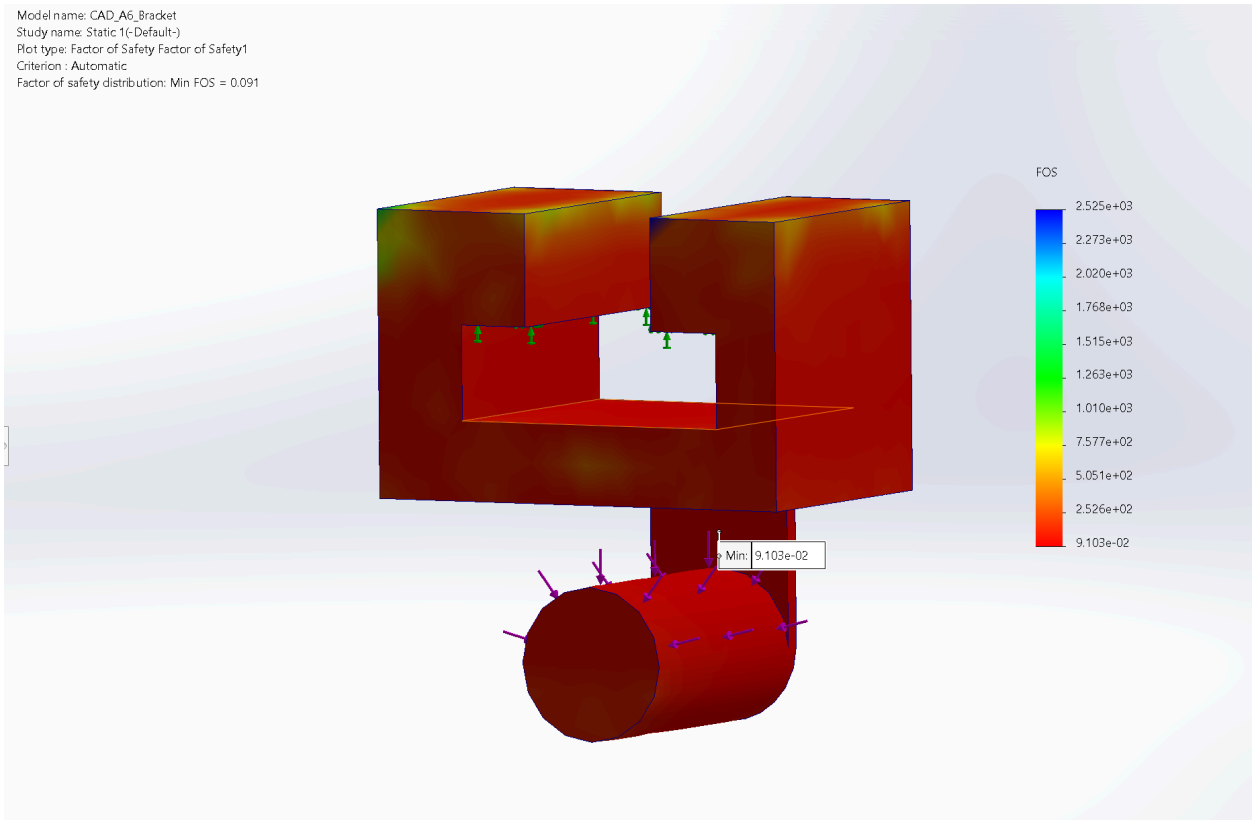
SOLIDWORKS Design 9020 0130 5187 4643 YWNH BJ4G

Coarse mesh:


Model name: CAD_A6_Bracket
Study name: Static 1(-Default-)
Plot type: Mesh Quality1
SOLIDWORKS Educational Product. For Instructional Use Only.

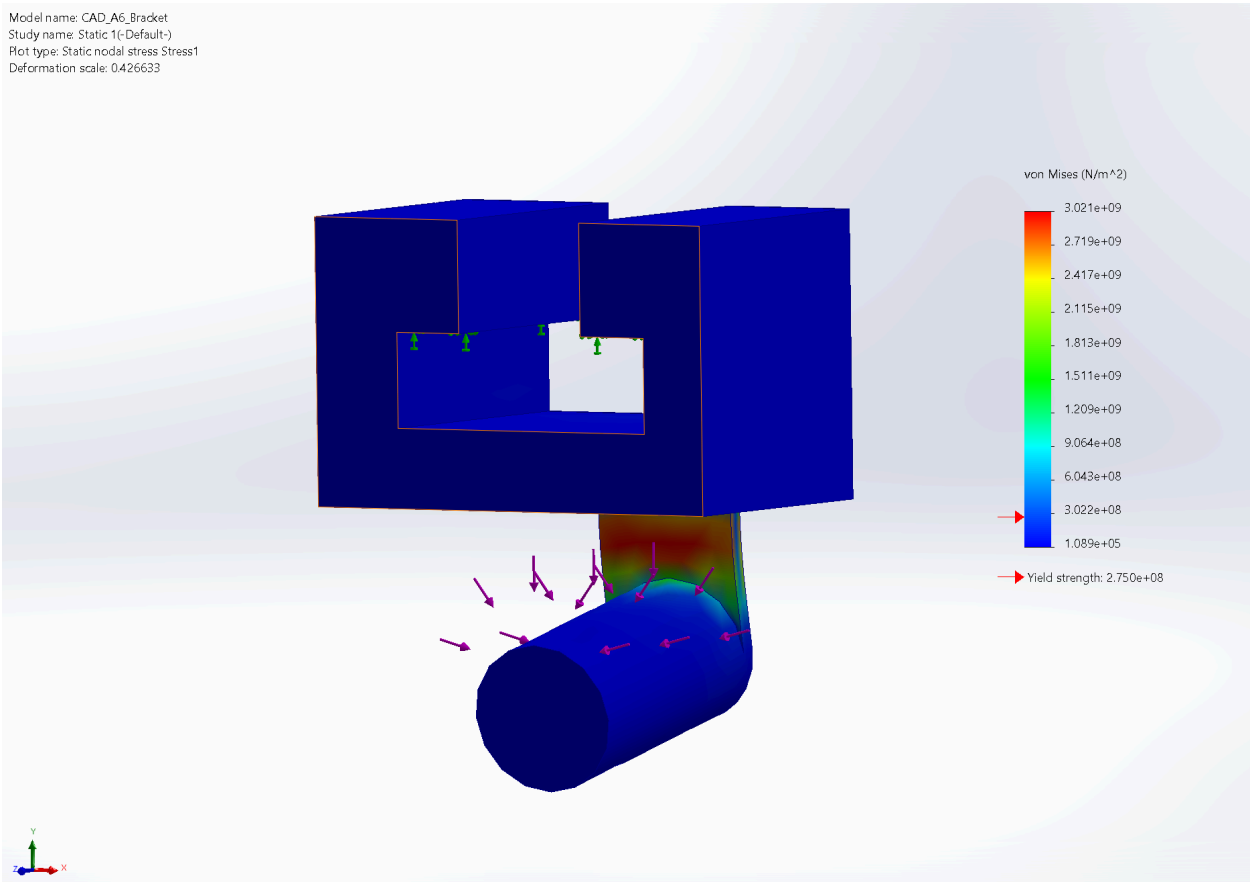


deflection

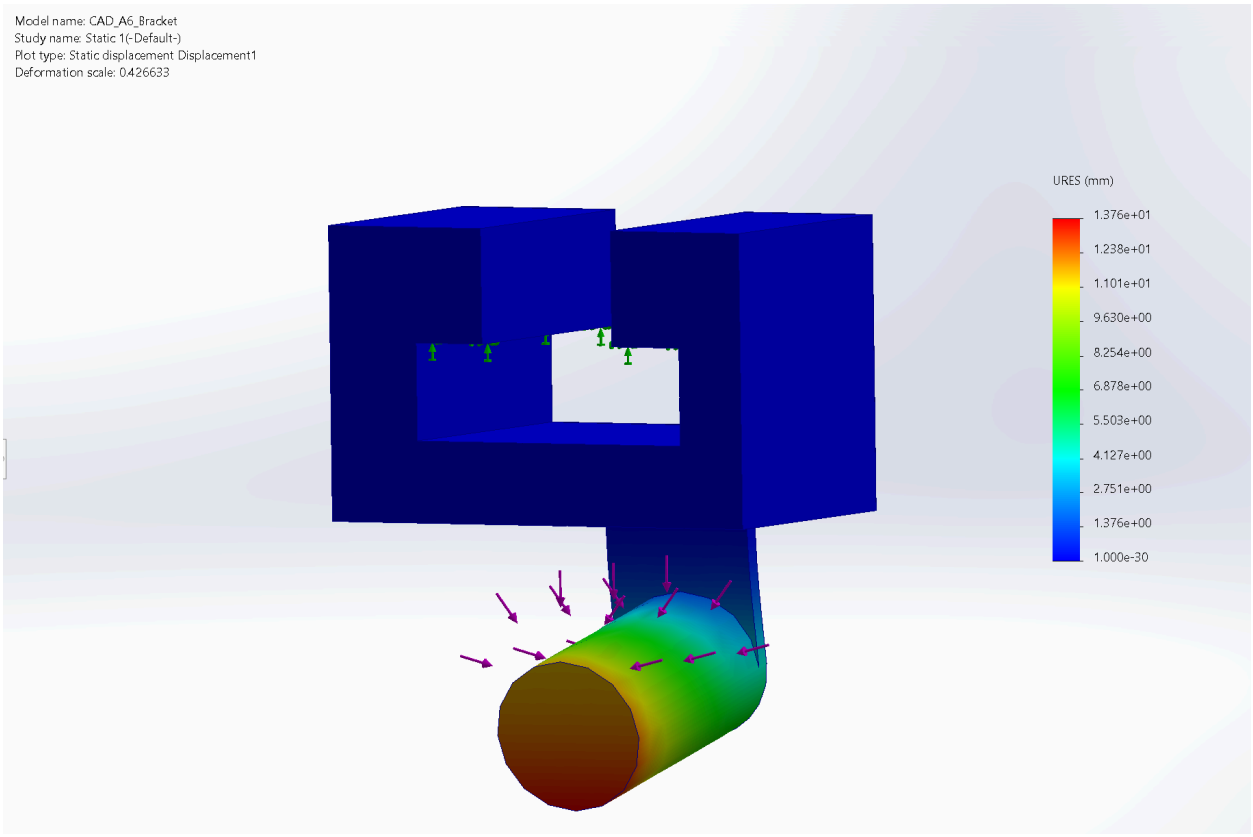


Factor of safety

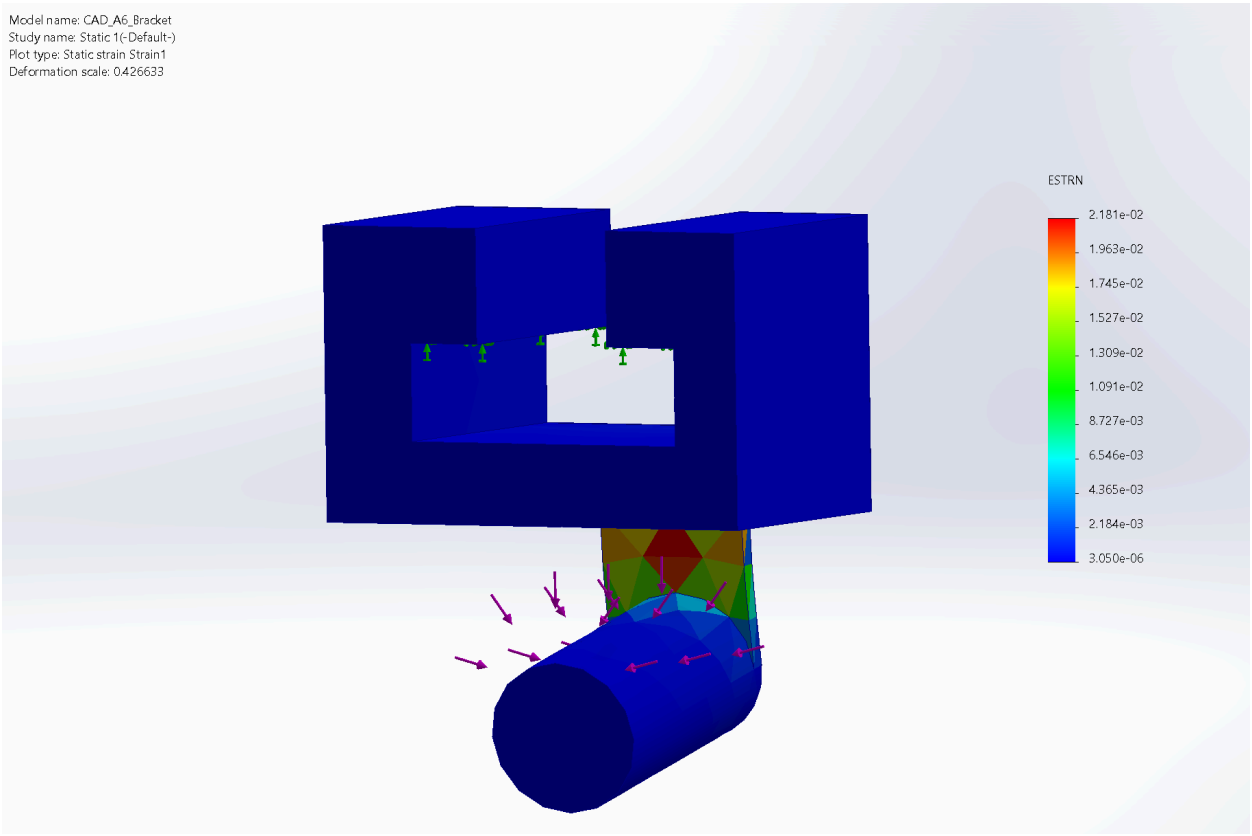
The minimum FOS, based on your defined material properties, is **0.0910**



Stress

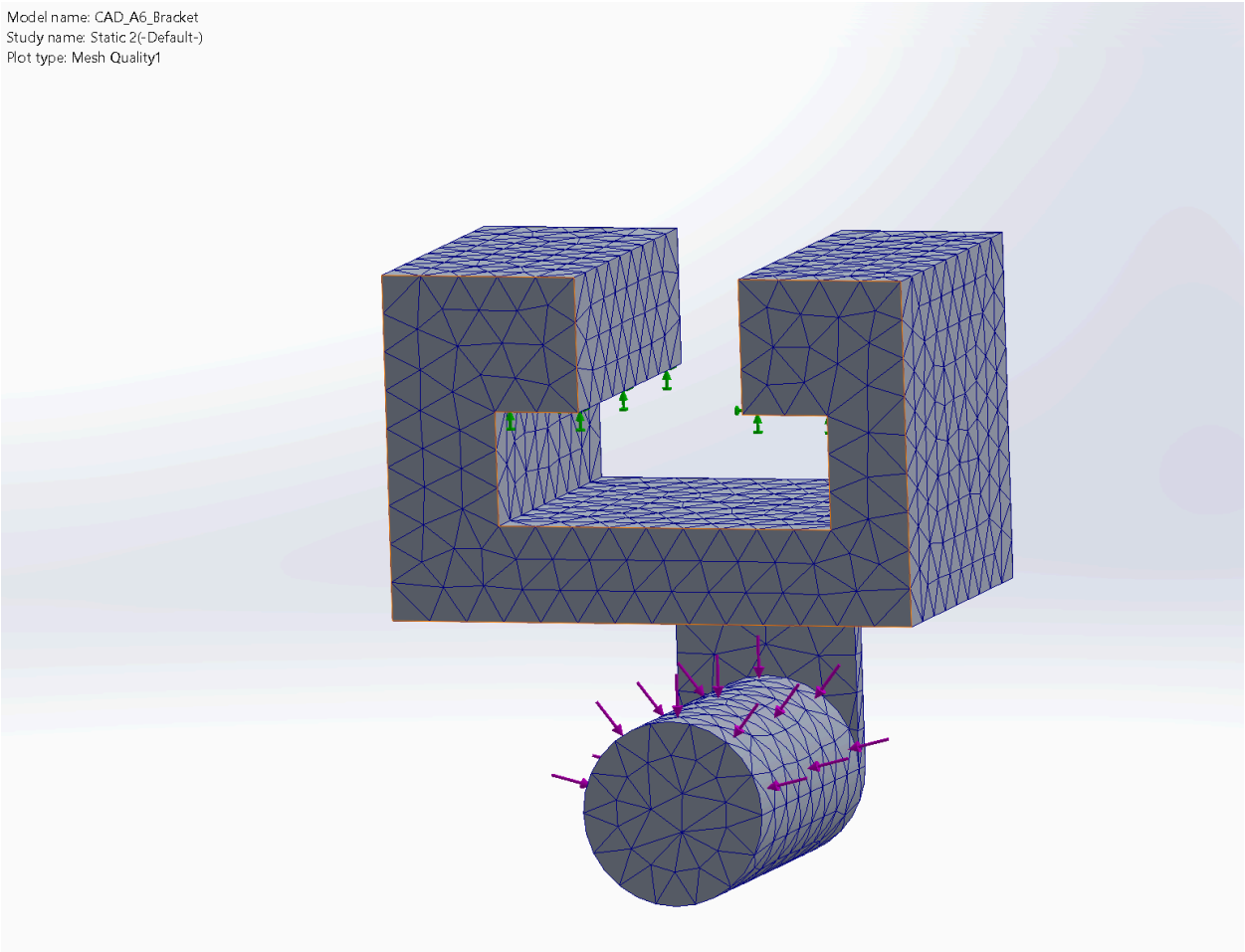


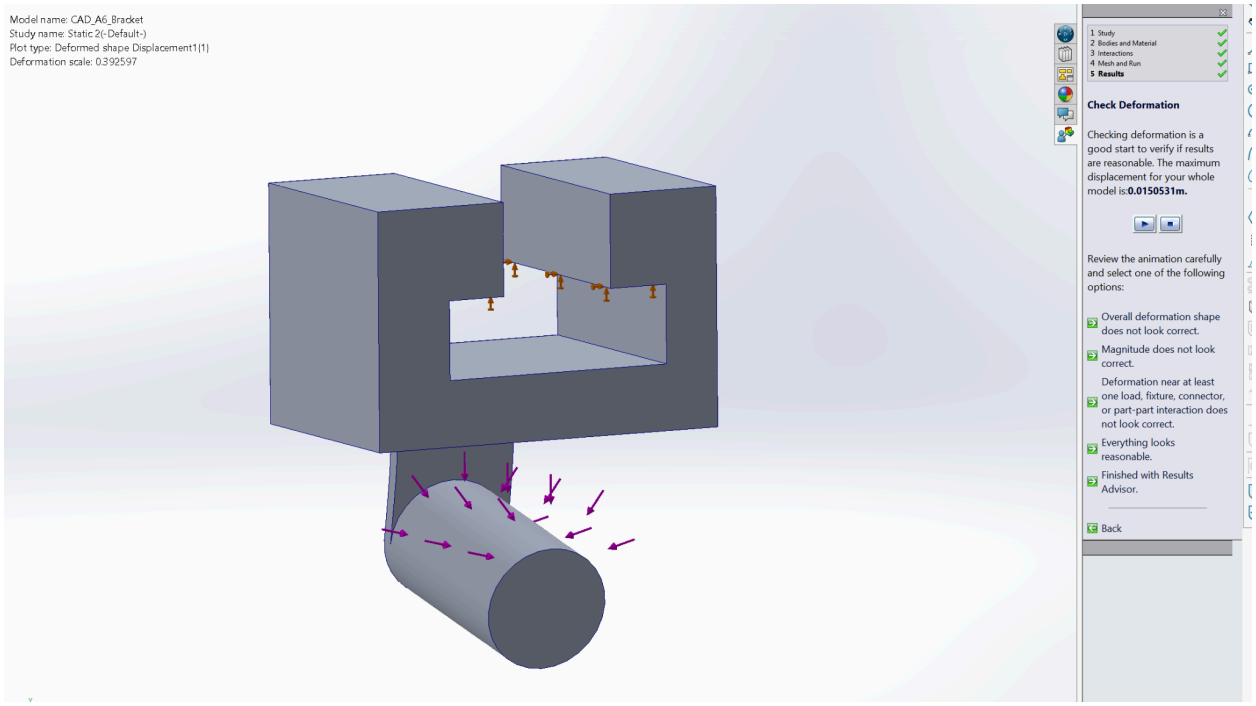
Displacement



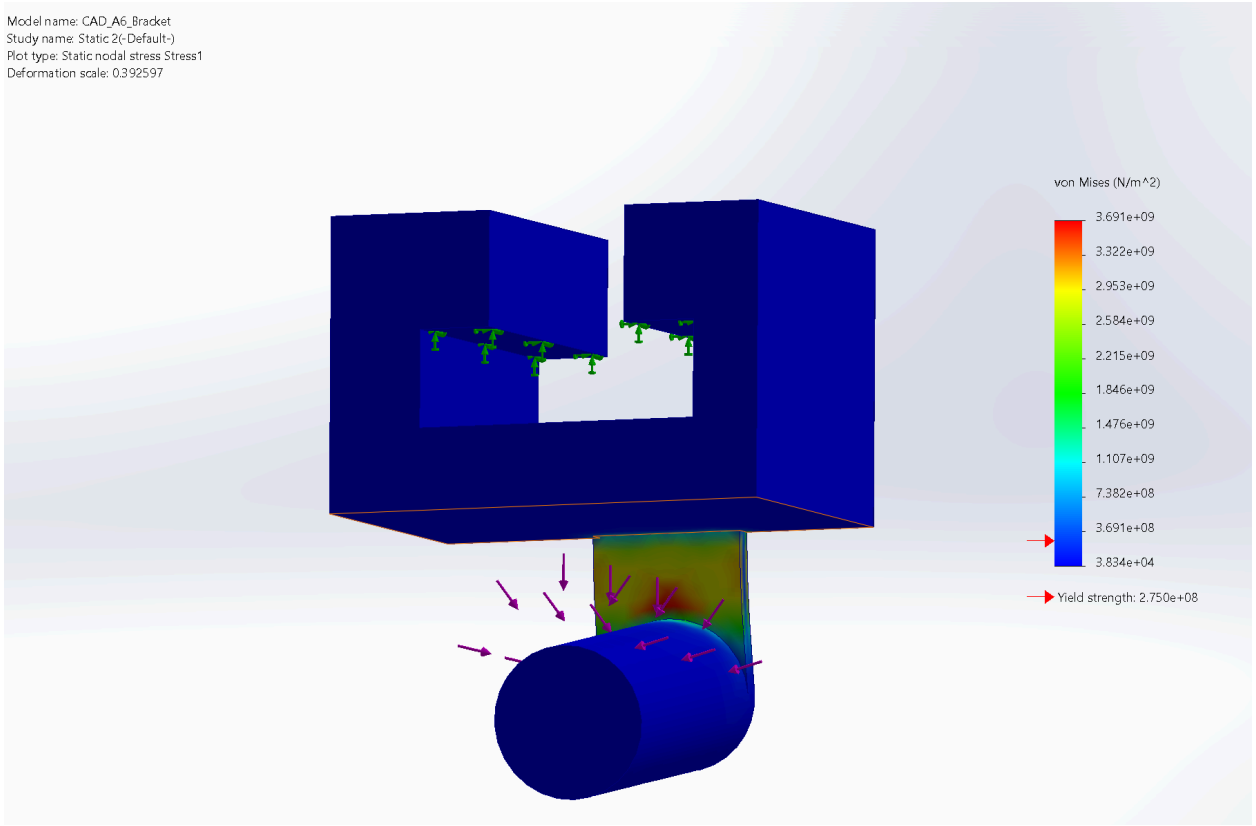
Strain

Medium Mesh:

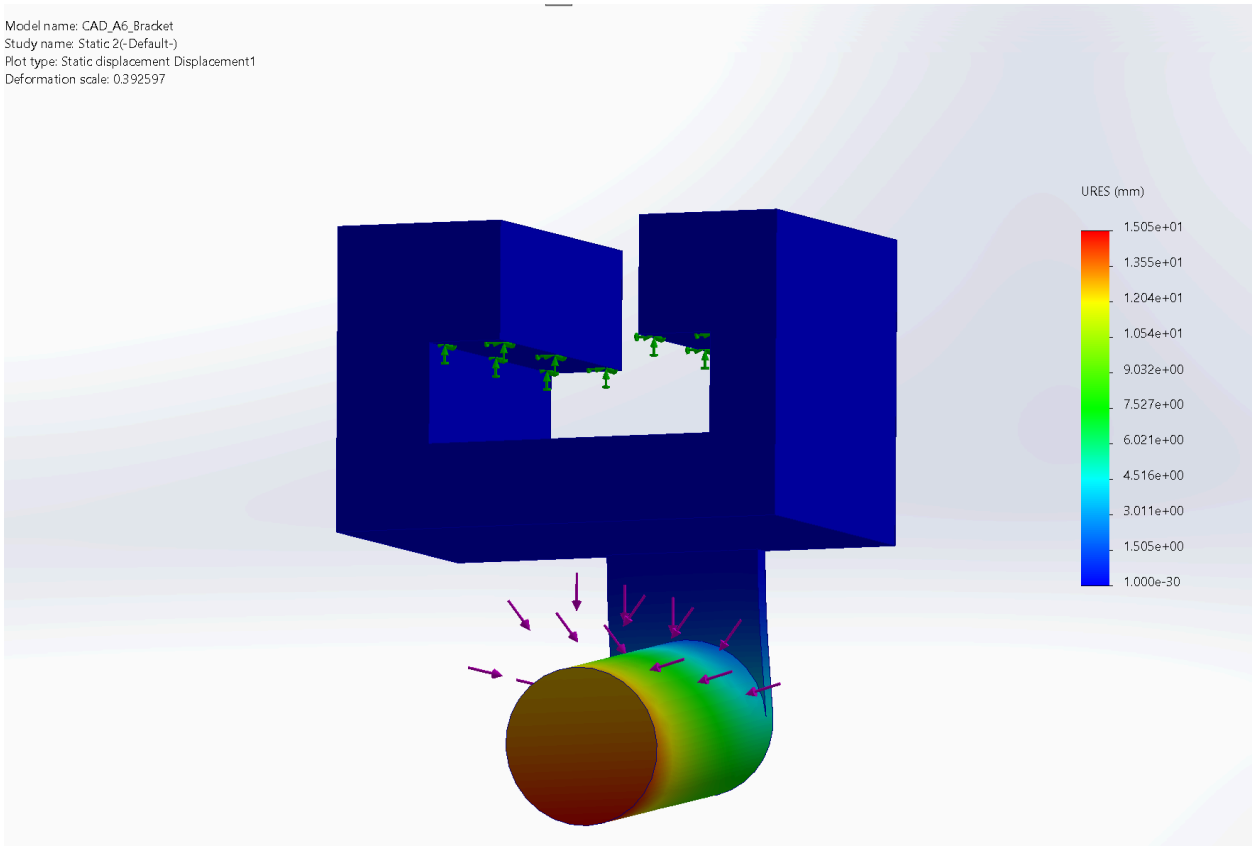




deformation



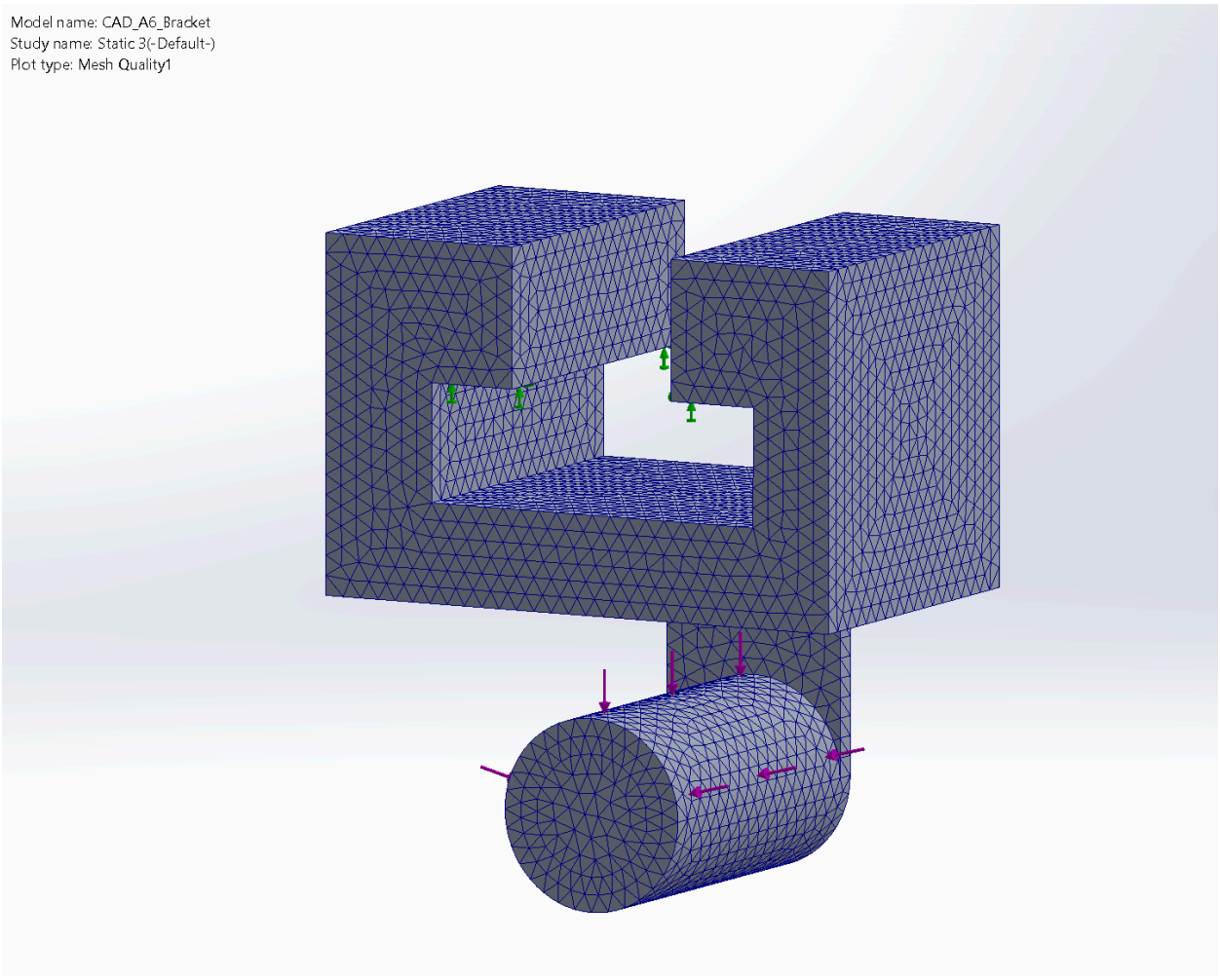
Stress

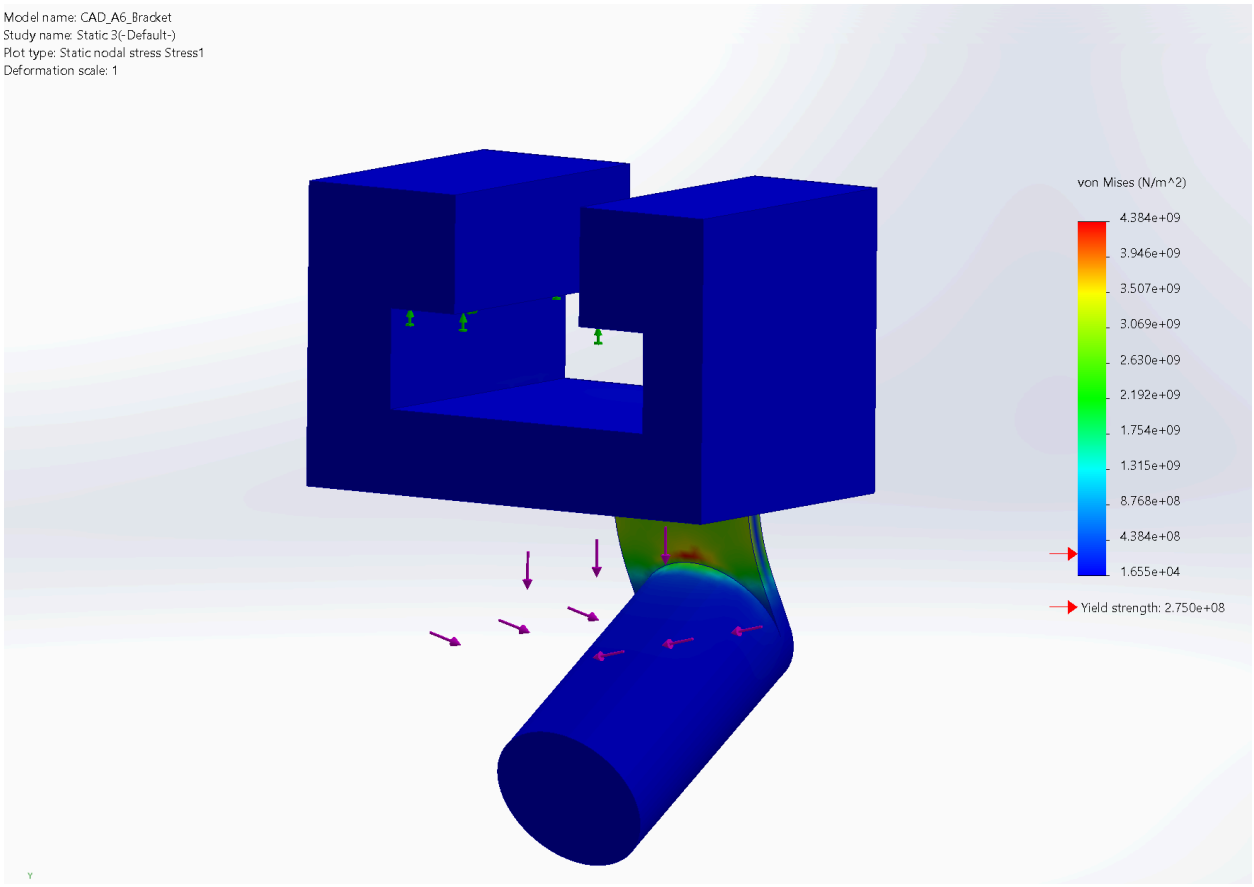


Deformation

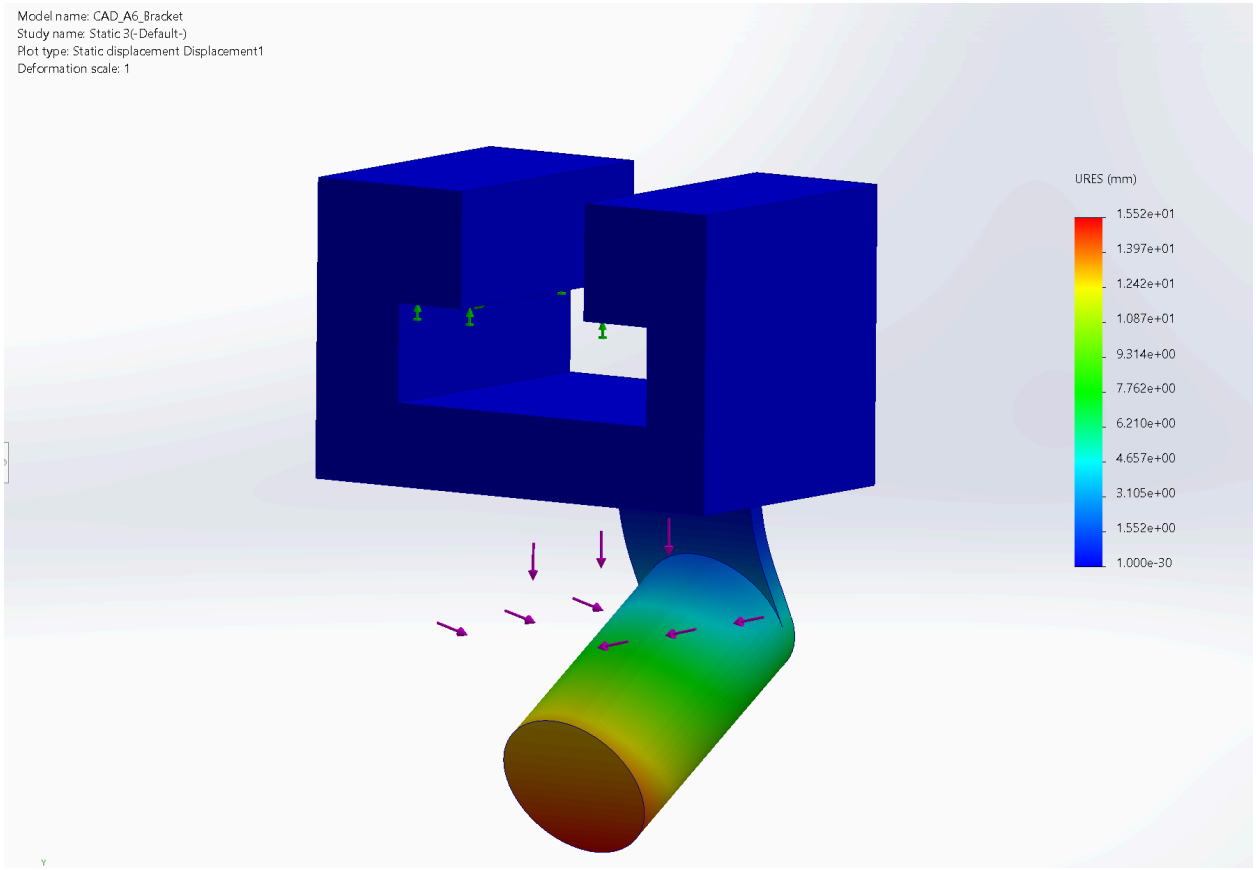
Fine mesh:

Model name: CAD_A6_Bracket
Study name: Static 3(-Default-)
Plot type: Mesh Quality1

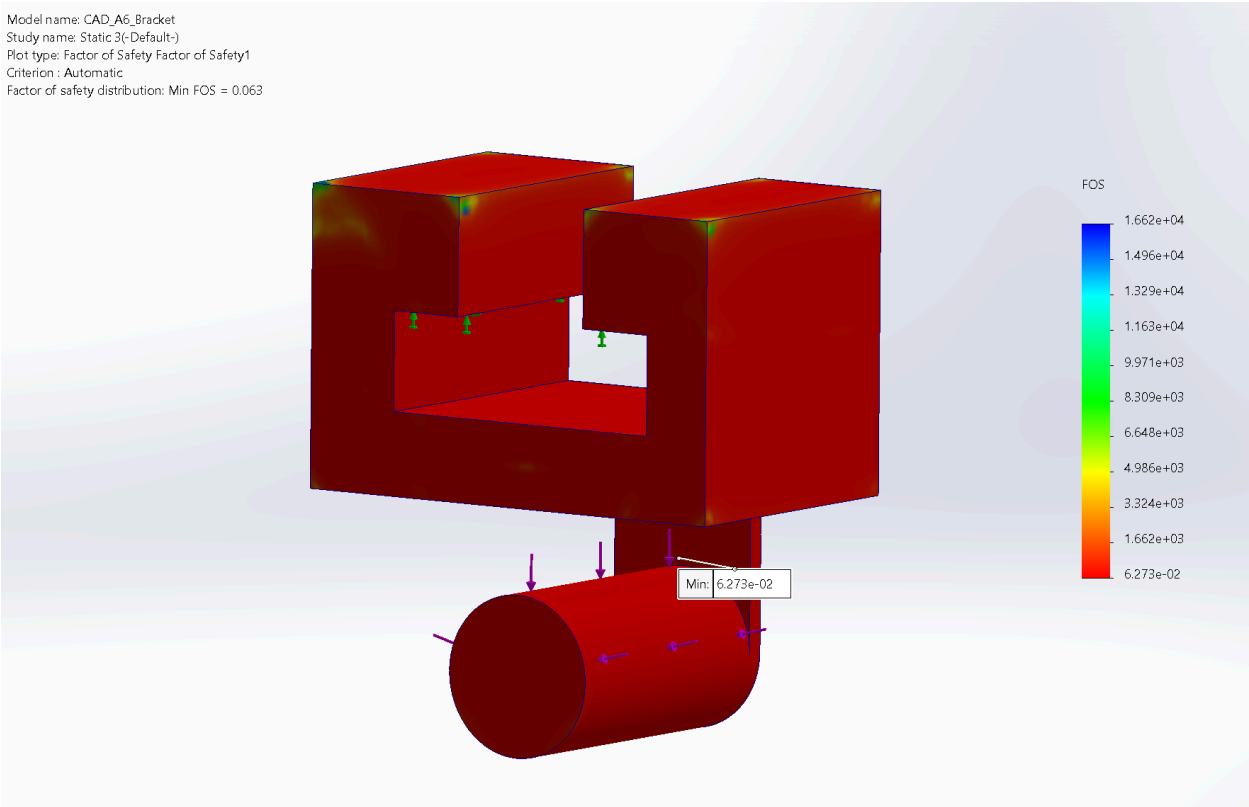



Model name: CAD_A6_Bracket
Study name: Static 3(-Default-)
Plot type: Static nodal stress Stress1
Deformation scale: 1
von Mises (N/m^2)
4.384e+09
3.946e+09
3.507e+09
3.069e+09
2.630e+09
2.192e+09
1.754e+09
1.315e+09
8.768e+08
4.384e+08
1.655e+04
Yield strength: 2.750e+08

Stress

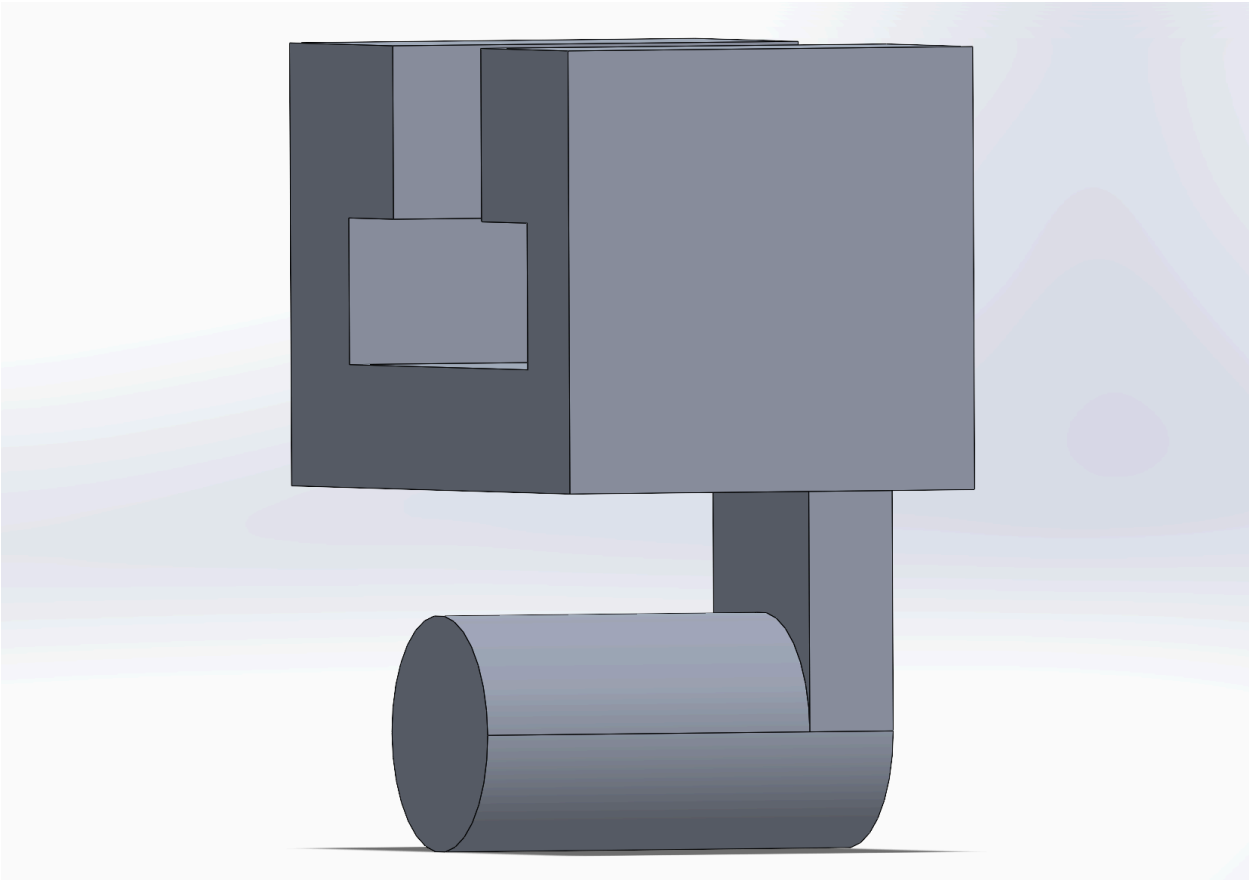

Model name: CAD_A6_Bracket
Study name: Static 3(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 1
URES (mm)
1.552e+01
1.397e+01
1.242e+01
1.087e+01
9.314e+00
7.762e+00
6.210e+00
4.657e+00
3.105e+00
1.552e+00
1.000e-30

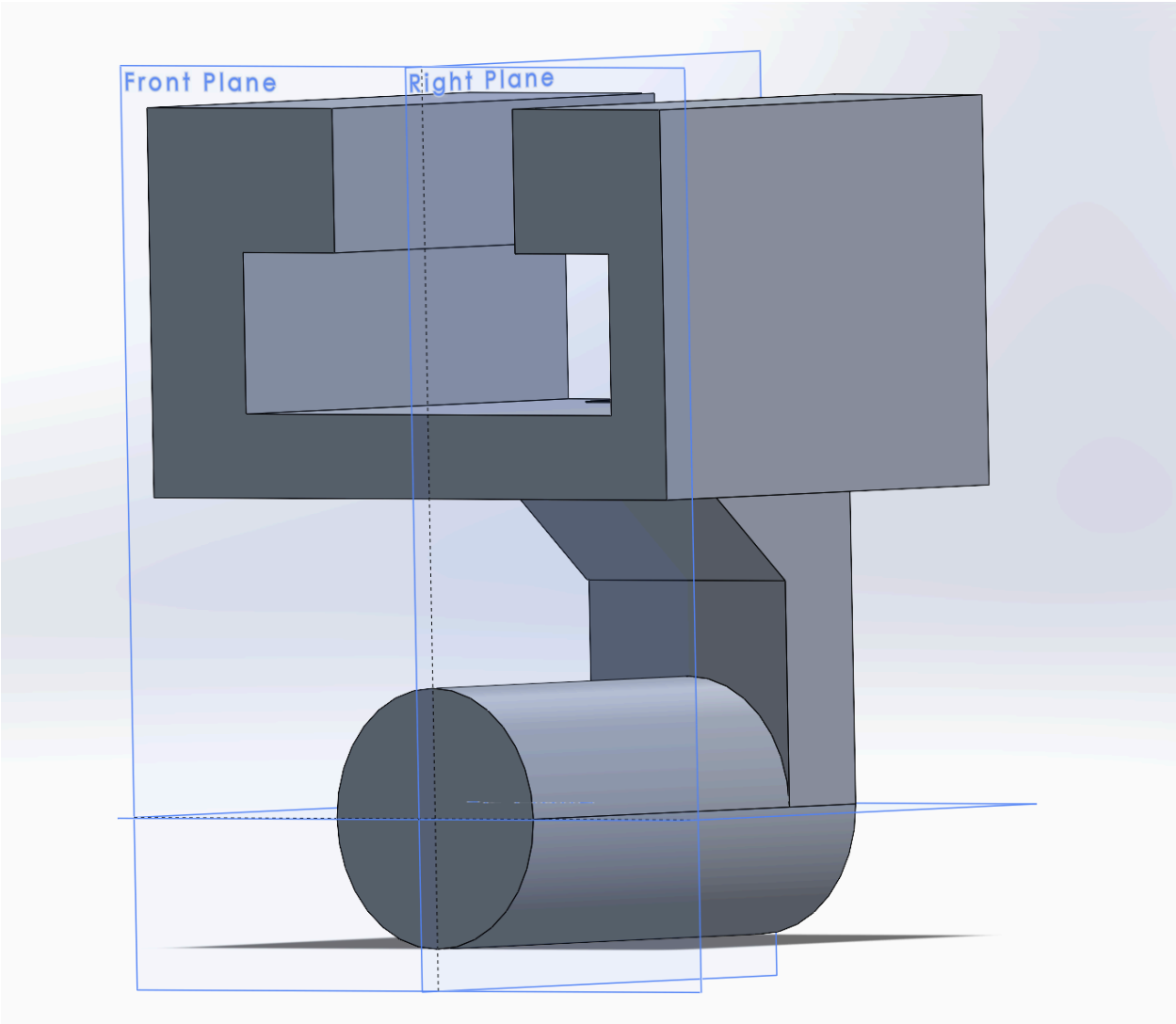
Deflection



Factor of Safety

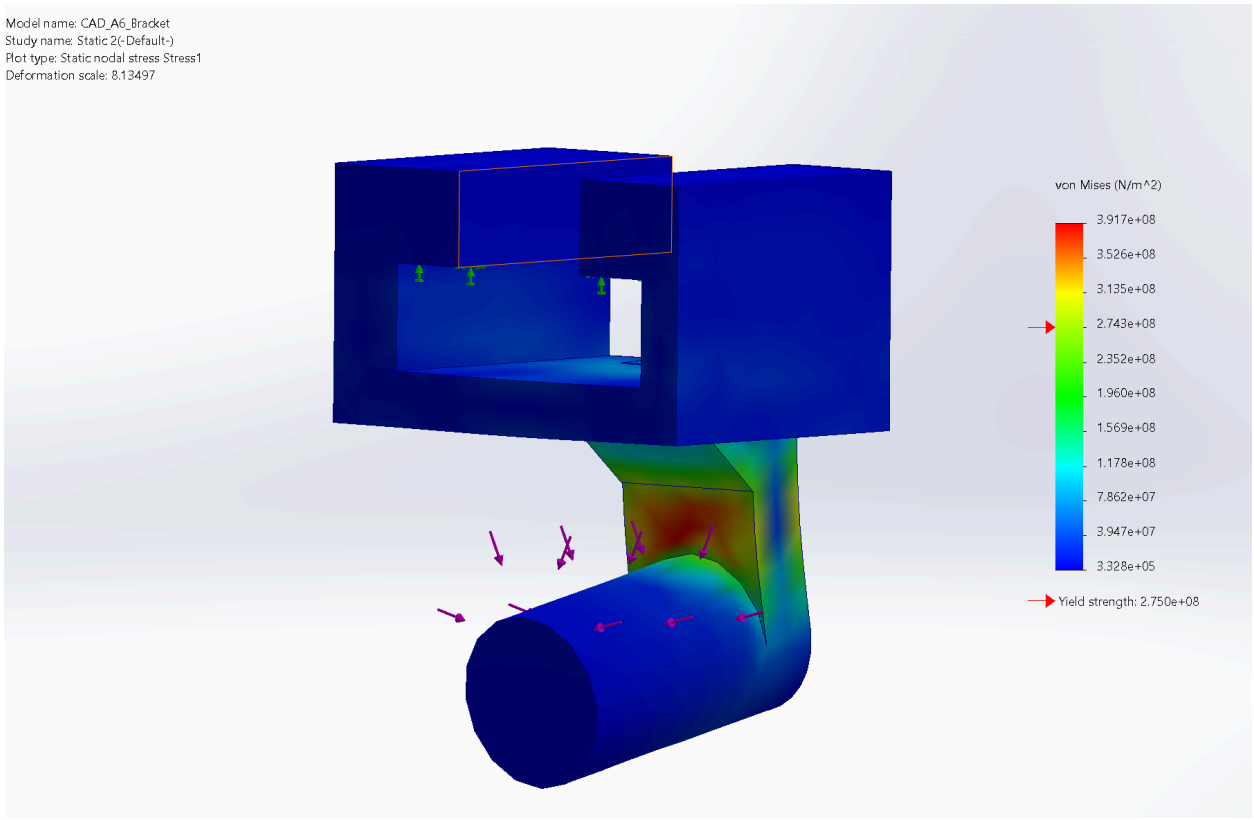
Design 2:



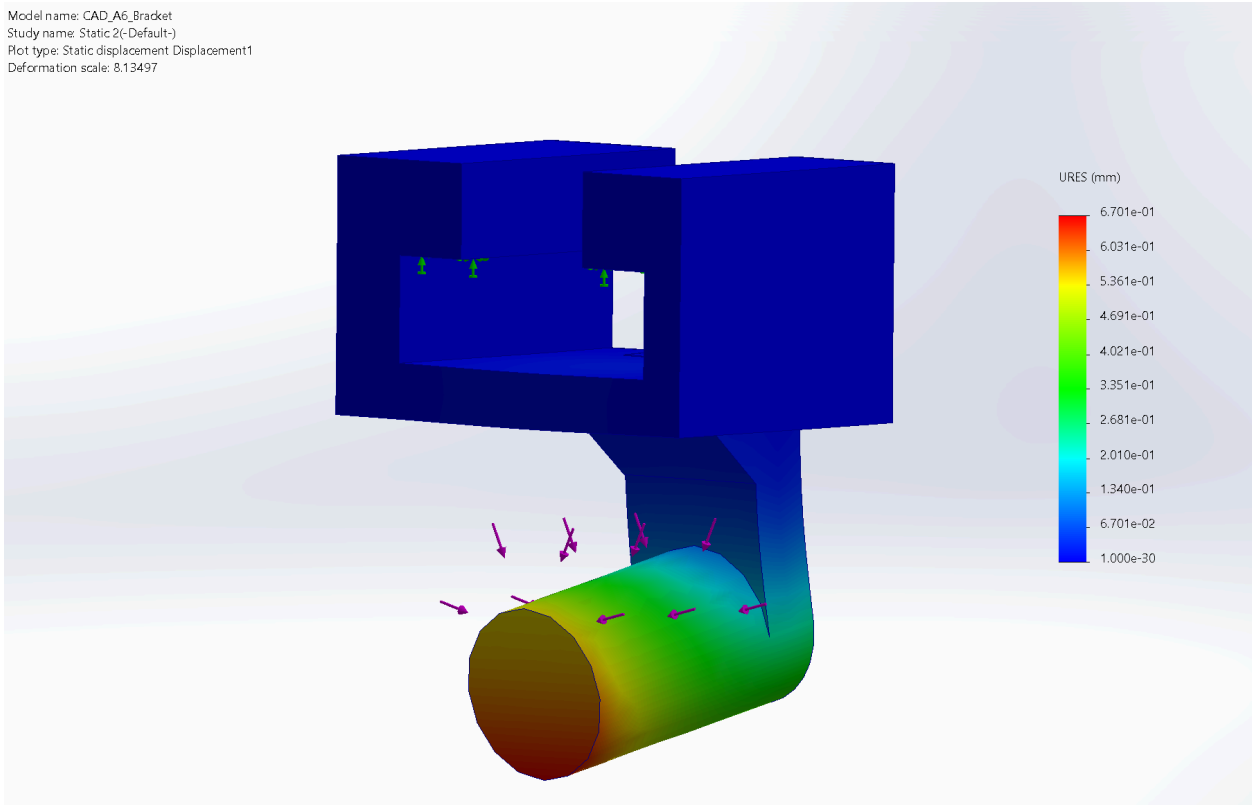


Shedded some weight on top and added it to the bottom to increase inertia, and resistance to stress

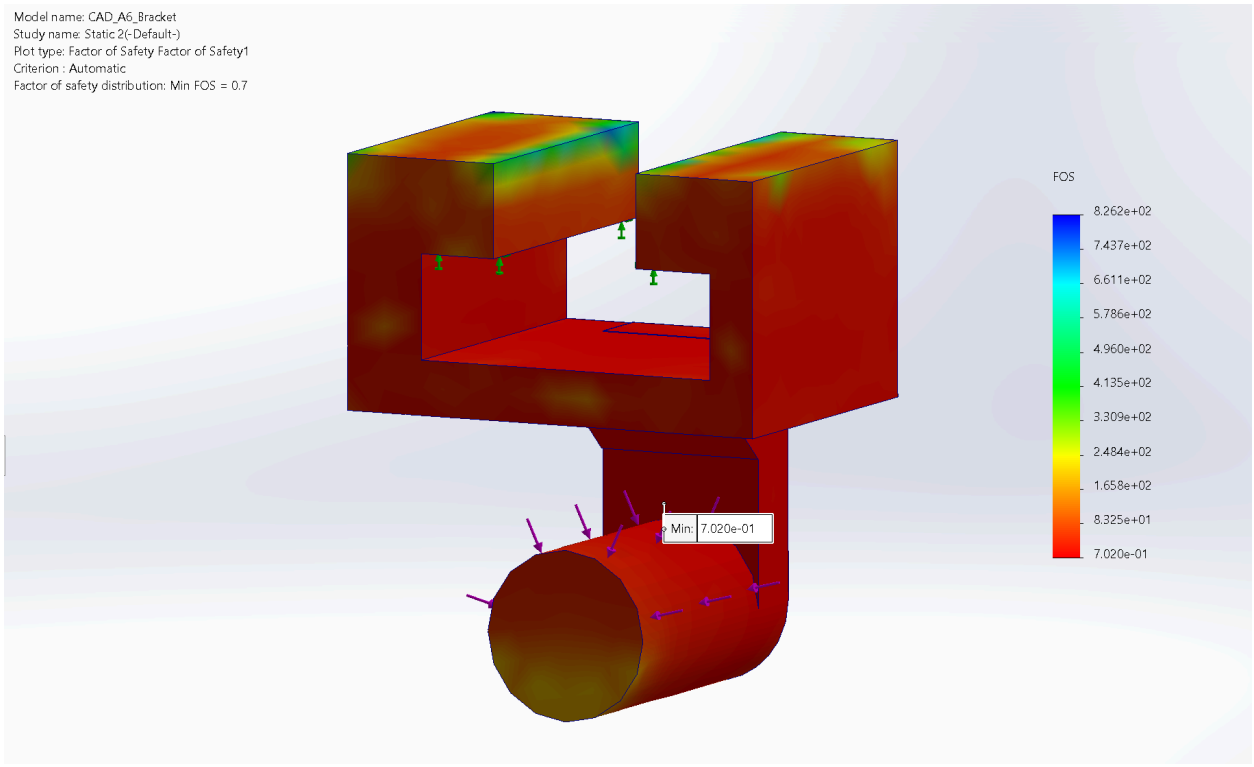
Stress:



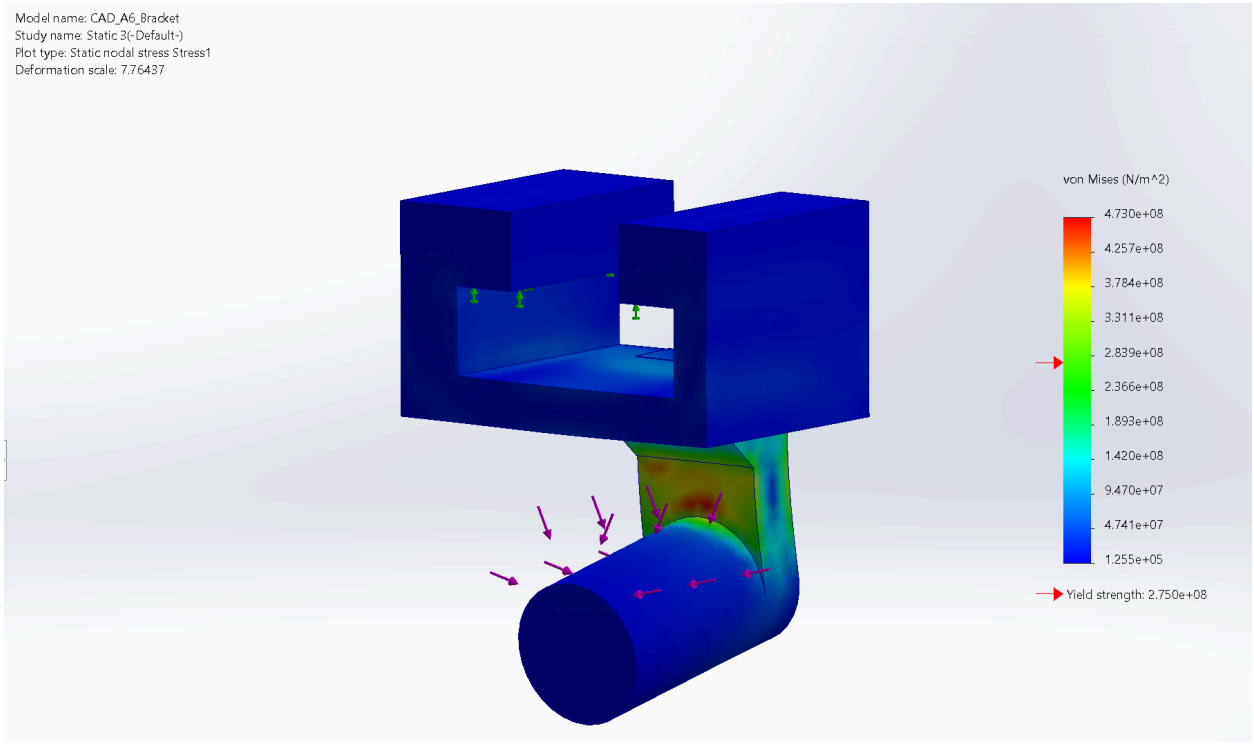
Deflection:



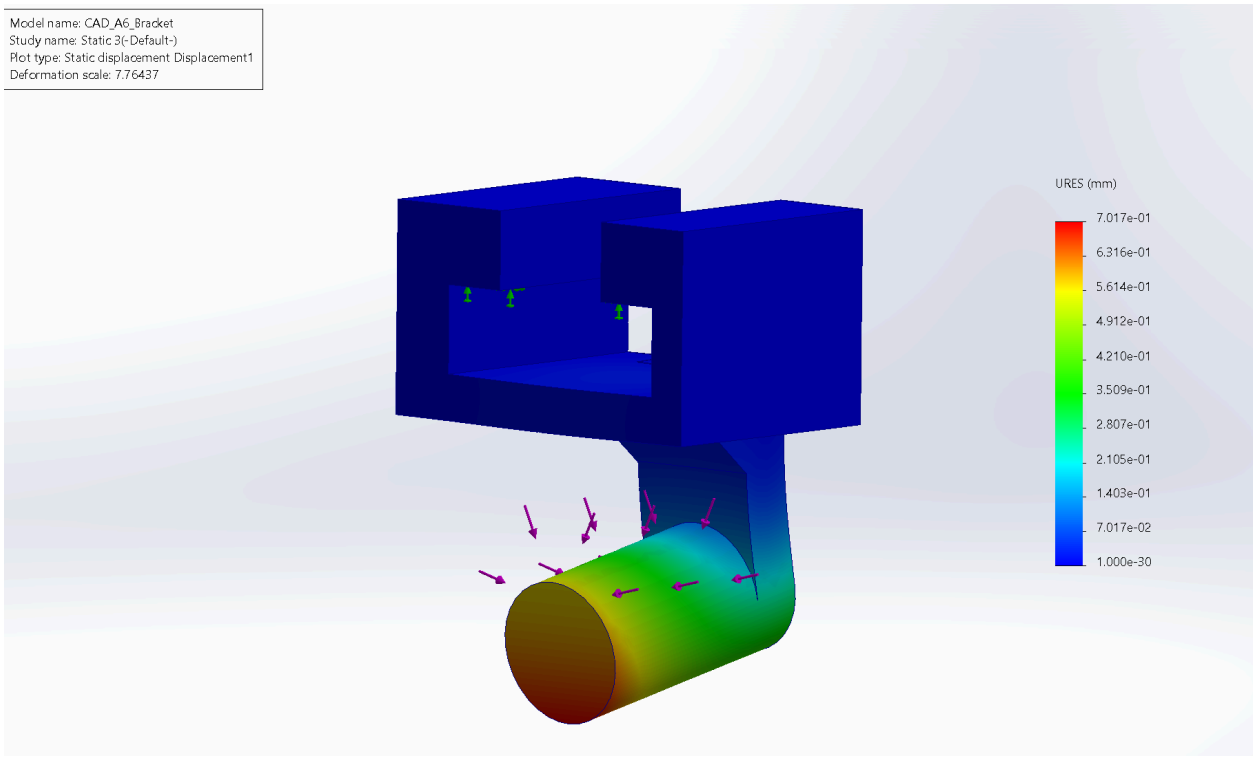
Factor of Safety:



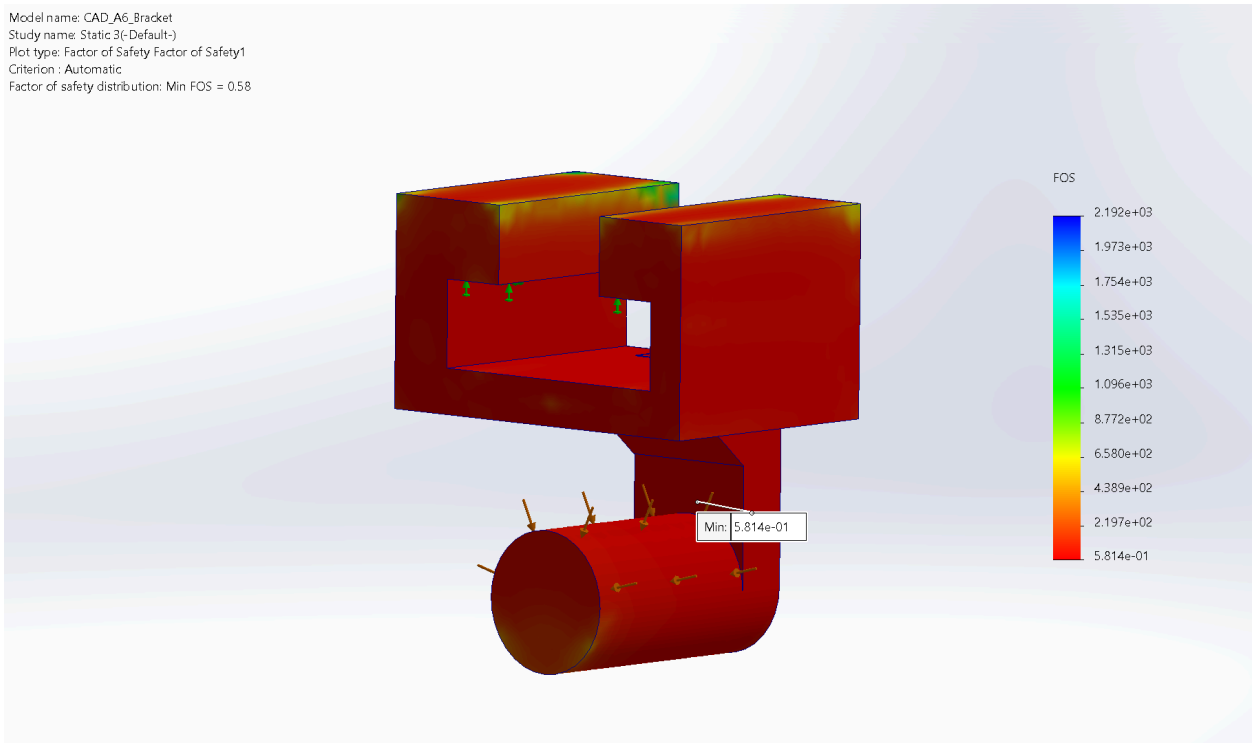
Medium coarse:



Stress

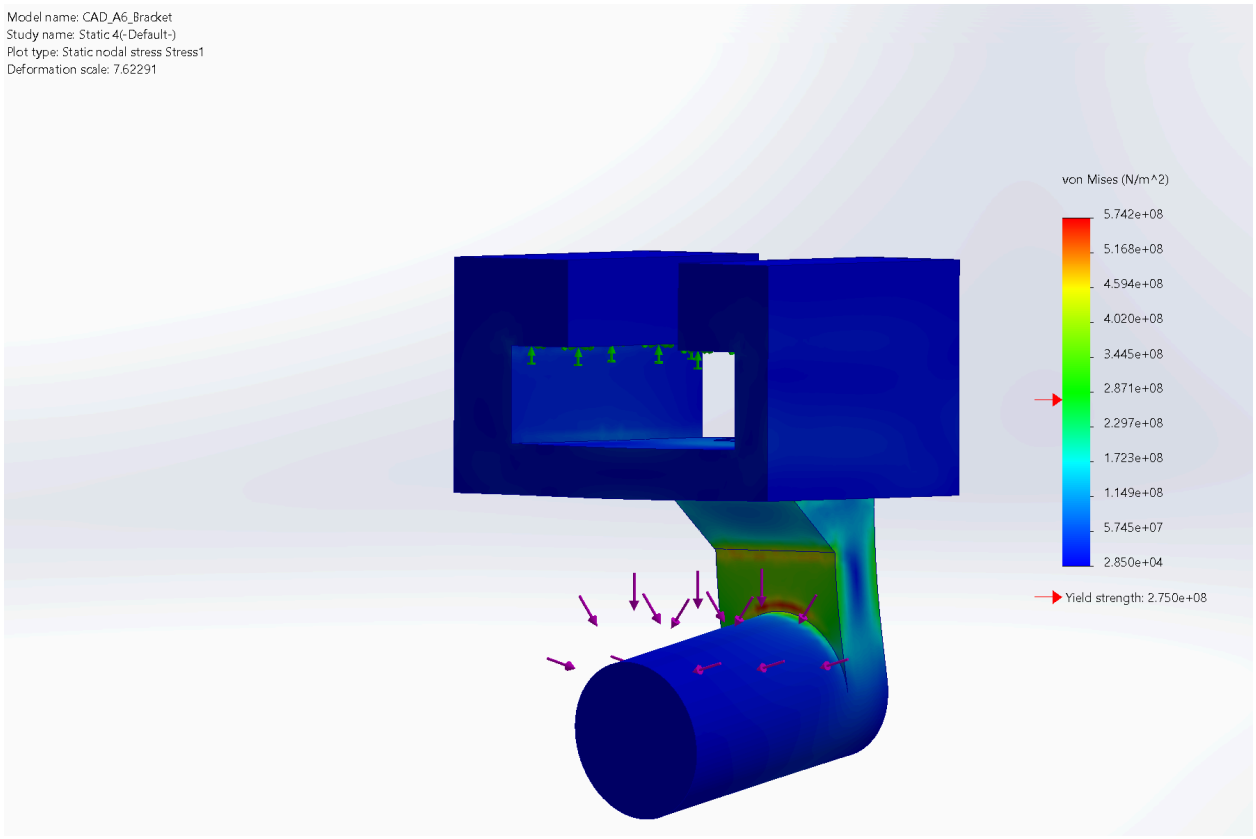


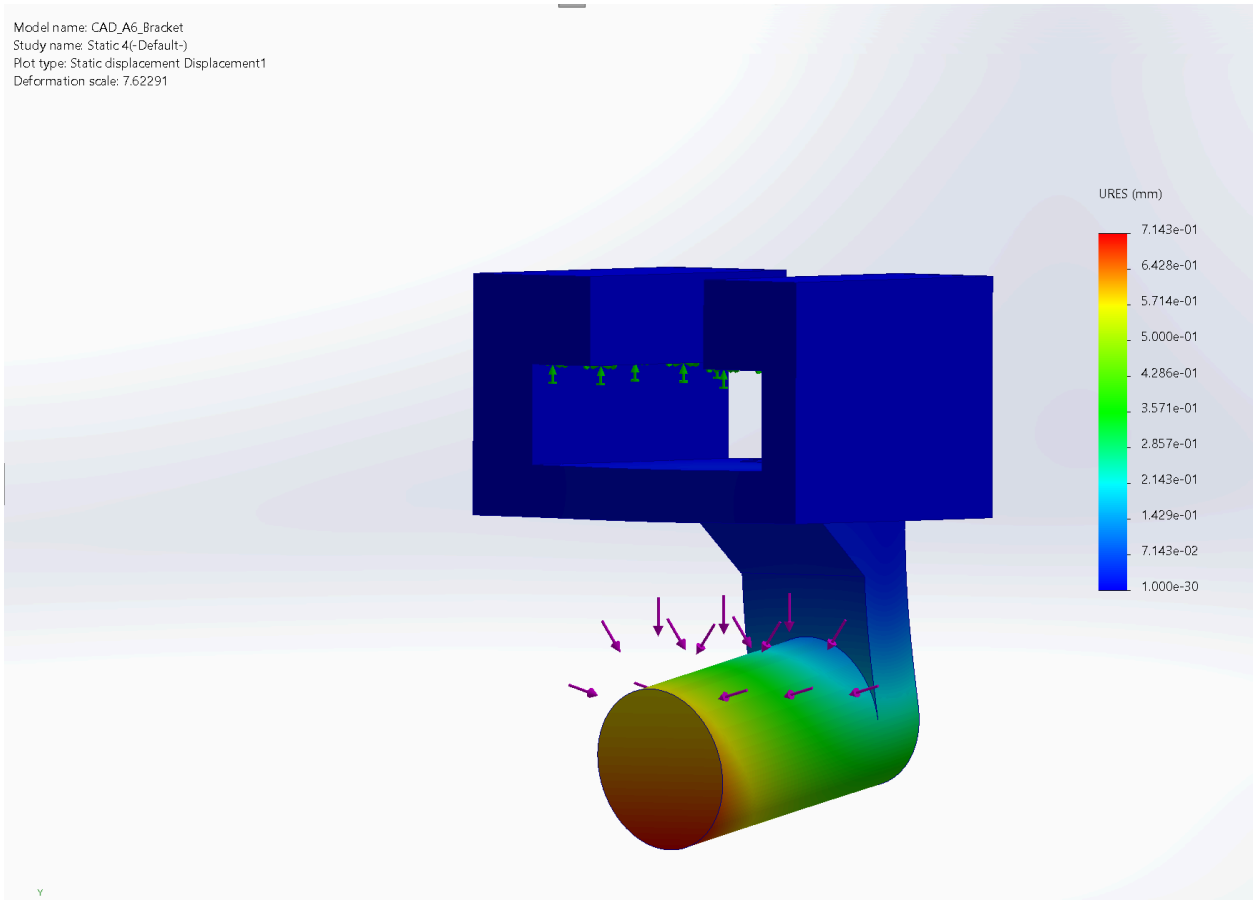
Deflection



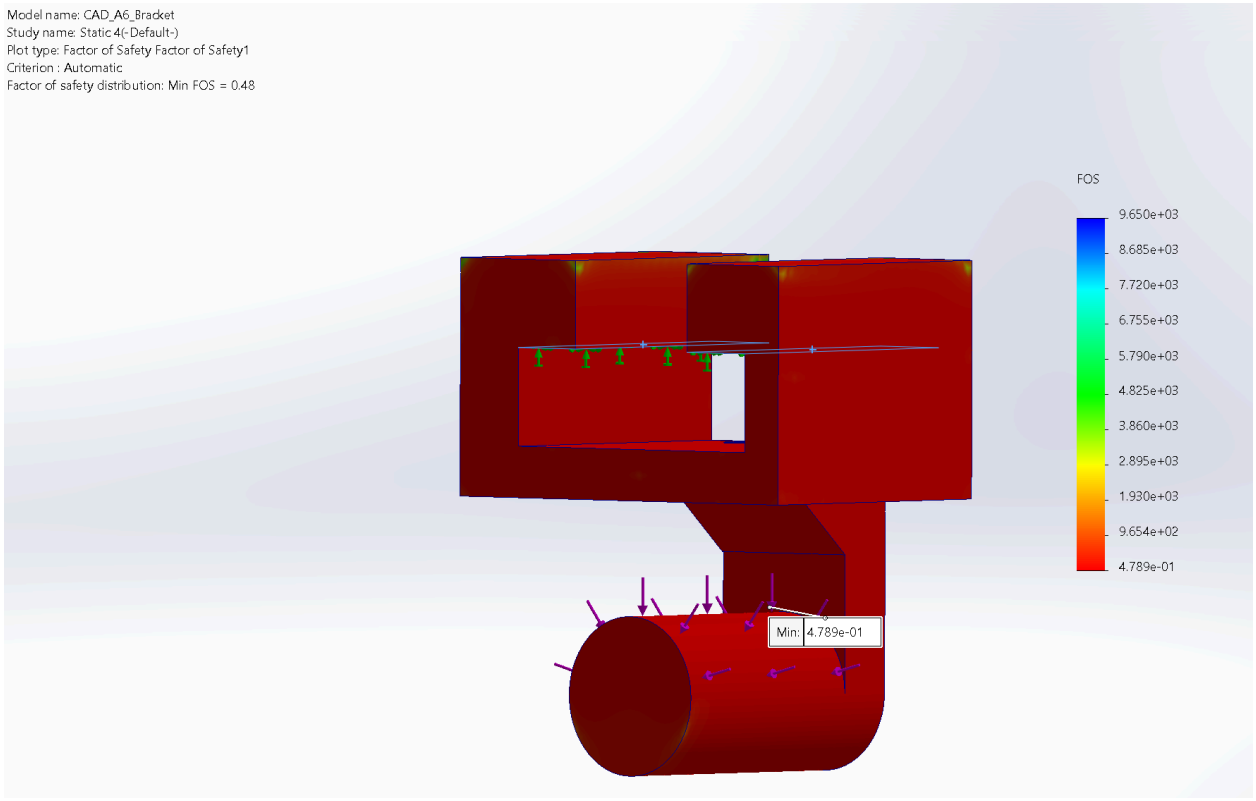
Factor of safety

Fine mesh:


Model name: CAD_A6_Bracket
Study name: Static 4(-Default-)
Plot type: Static nodal stress Stress1
Deformation scale: 7.62291
von Mises (N/m^2)
5.742e+08
5.168e+08
4.594e+08
4.020e+08
3.445e+08
2.871e+08
2.297e+08
1.723e+08
1.149e+08
5.745e+07
2.850e+04
Yield strength: 2.750e+08


Model name: CAD_A6_Bracket
Study name: Static 4(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 7.62291
URES (mm)
7.143e-01
6.428e-01
5.714e-01
5.000e-01
4.286e-01
3.571e-01
2.857e-01
2.143e-01
1.429e-01
7.143e-02
1.000e-30
Y

deflection

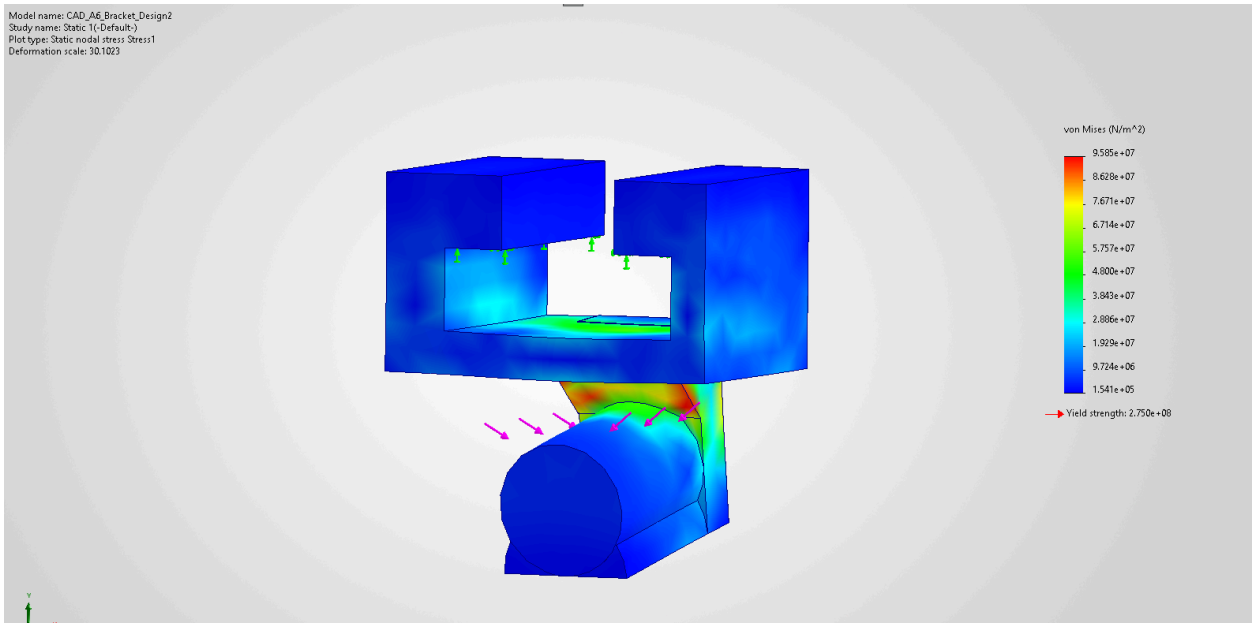


Factor of safety

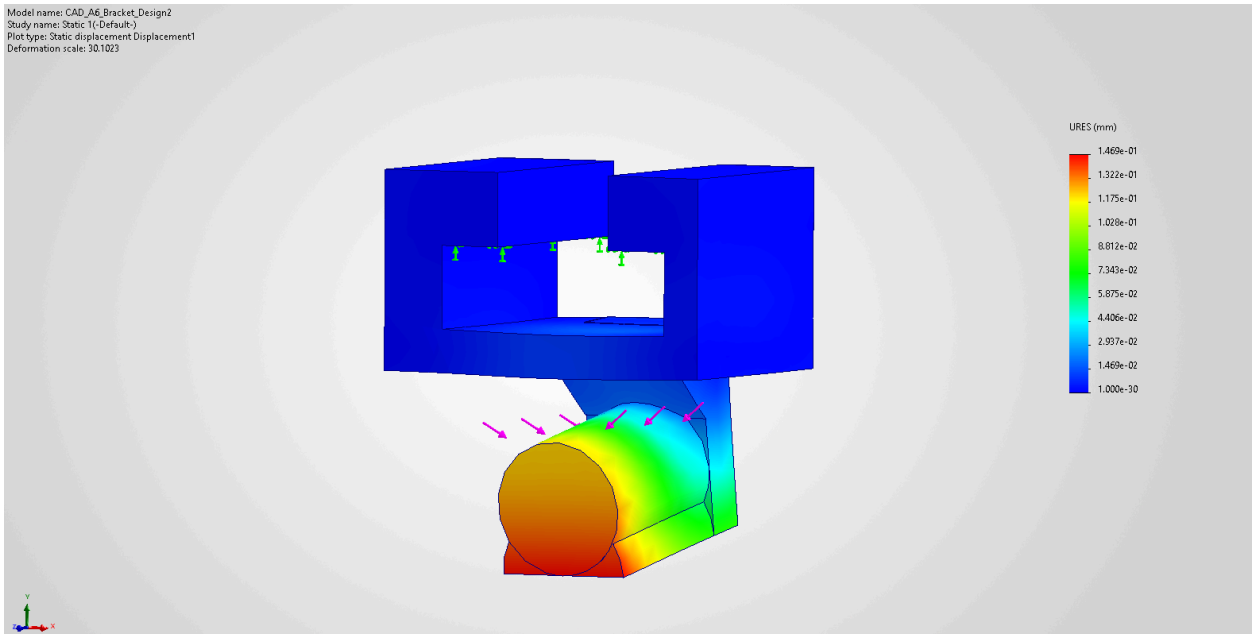
Design 3:

Goal increase factor of safety, reduce deflection, mass = 0.09 lbs

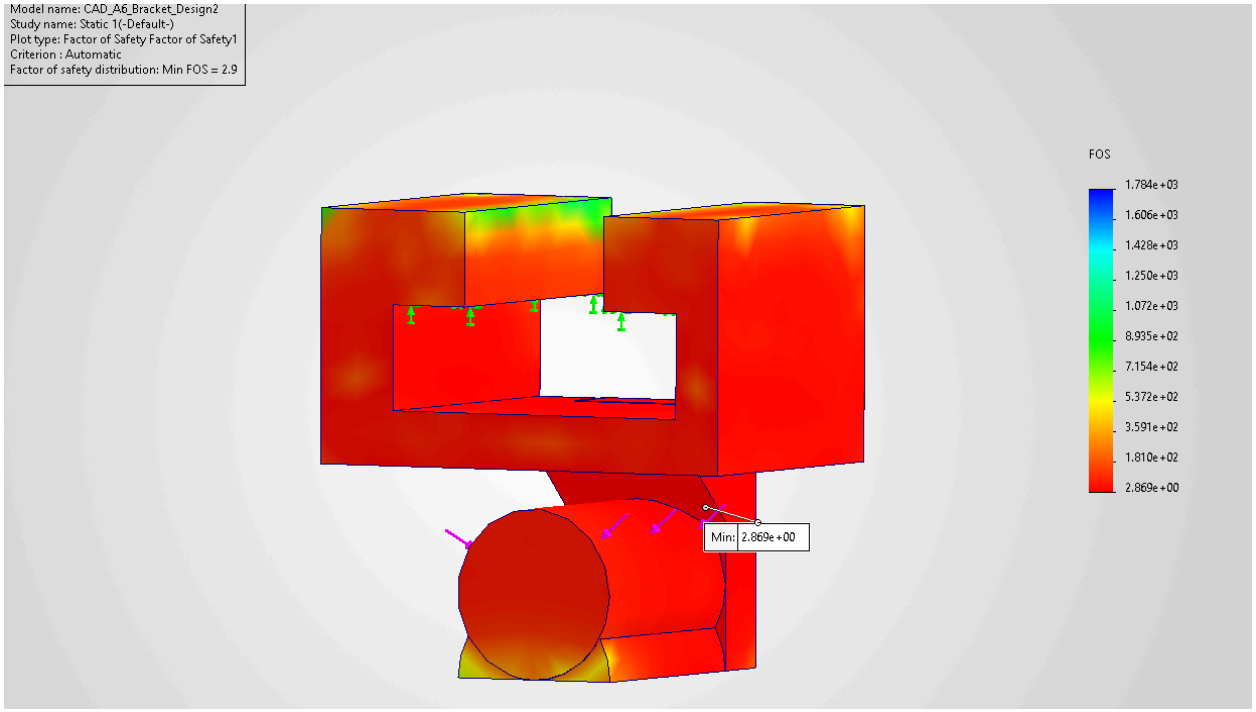
Coarse mesh:



Stress

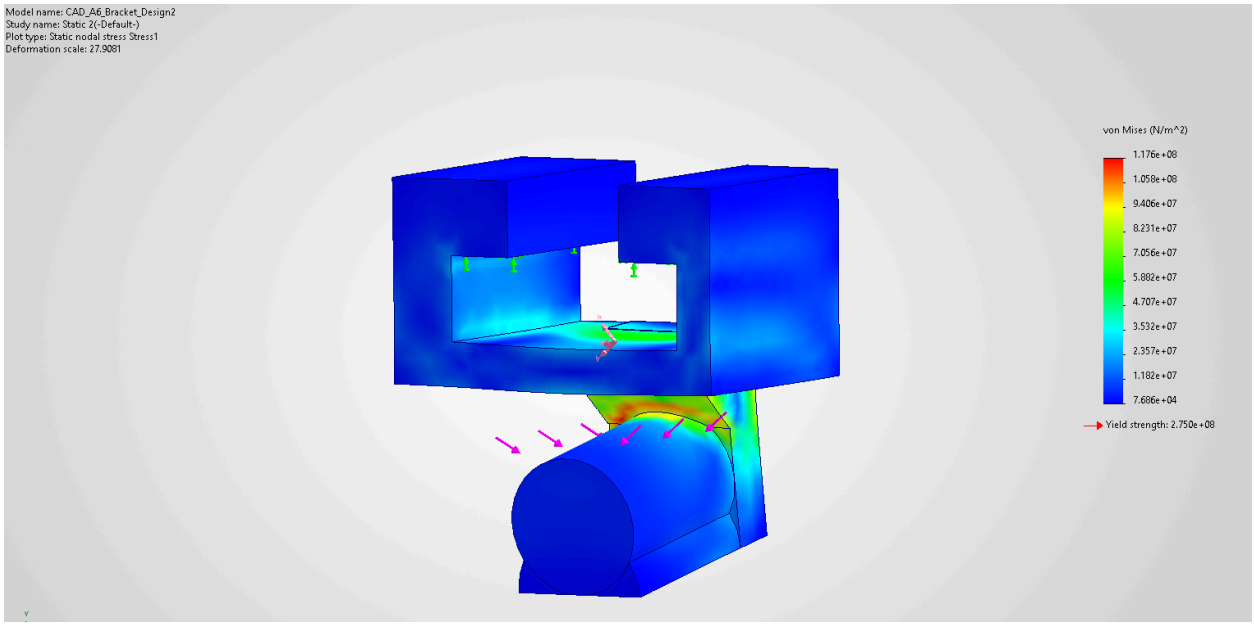


Deflection

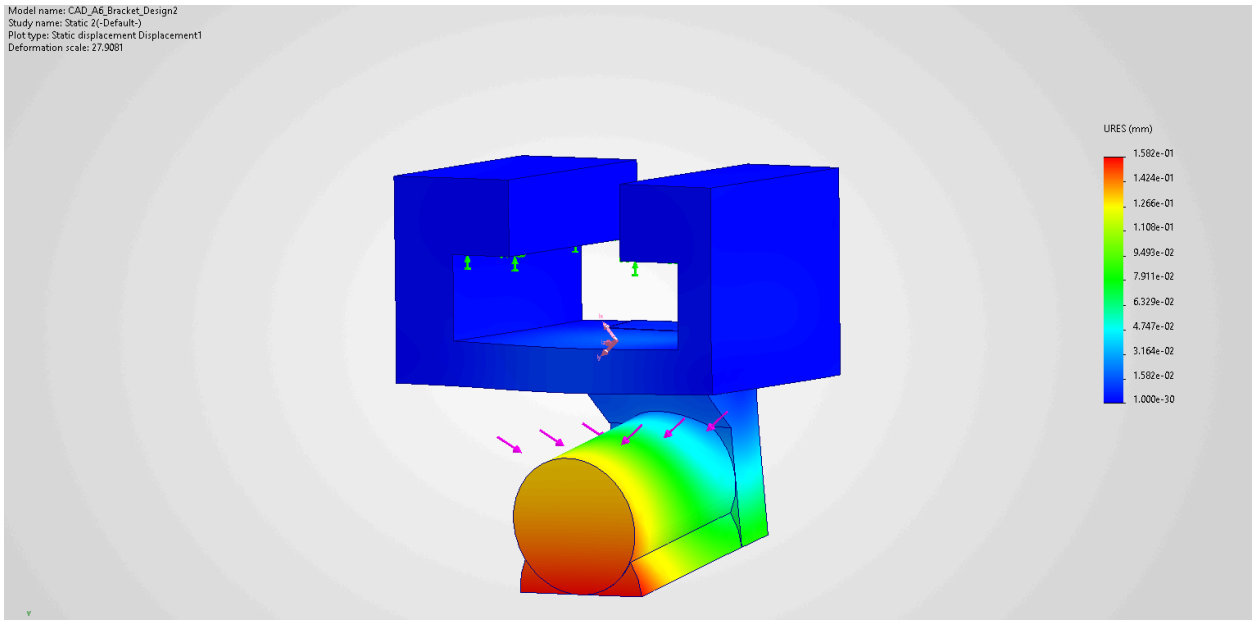


Factor of safety

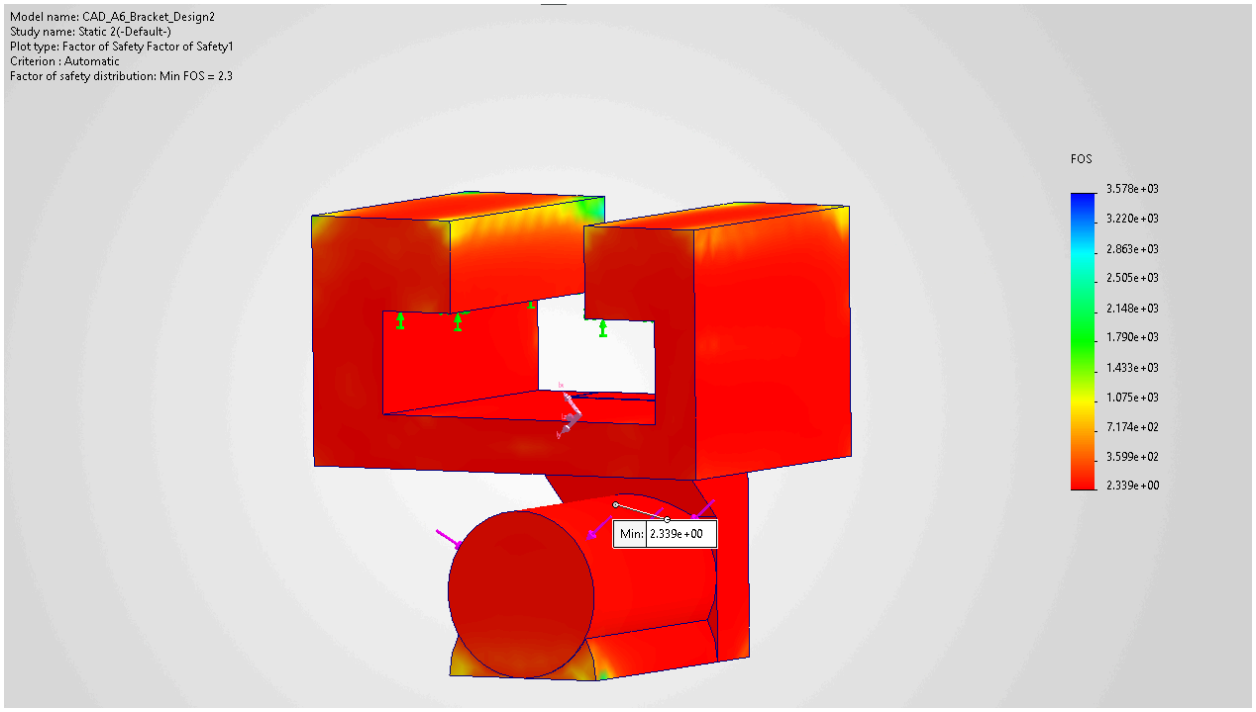
Medium mesh:



Stress

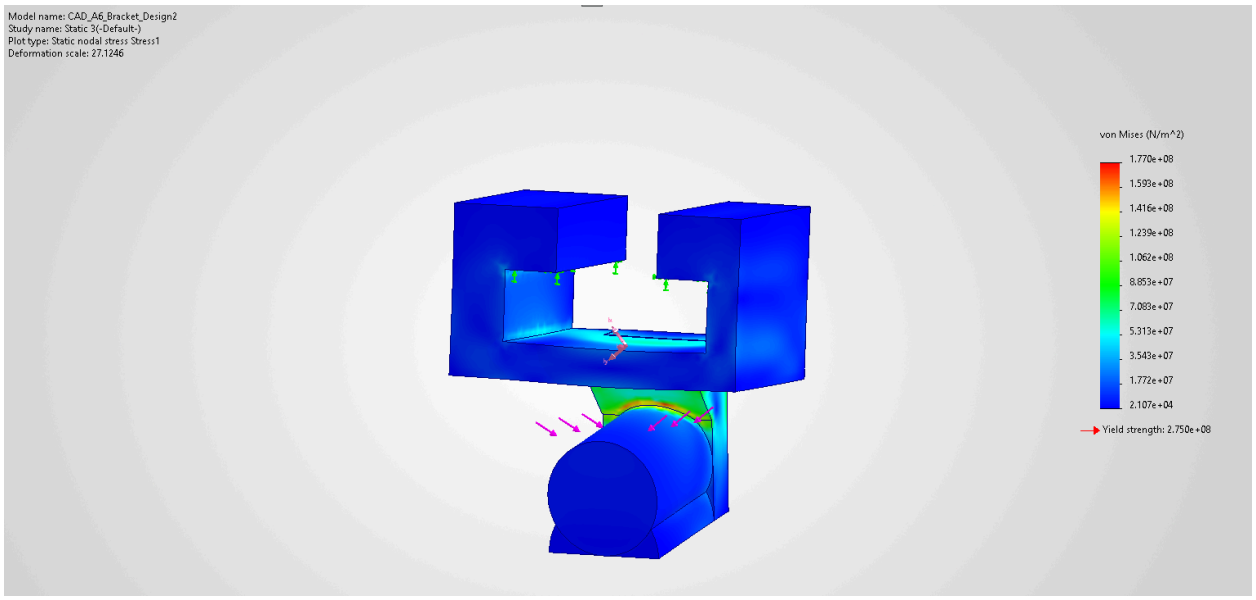


deflection

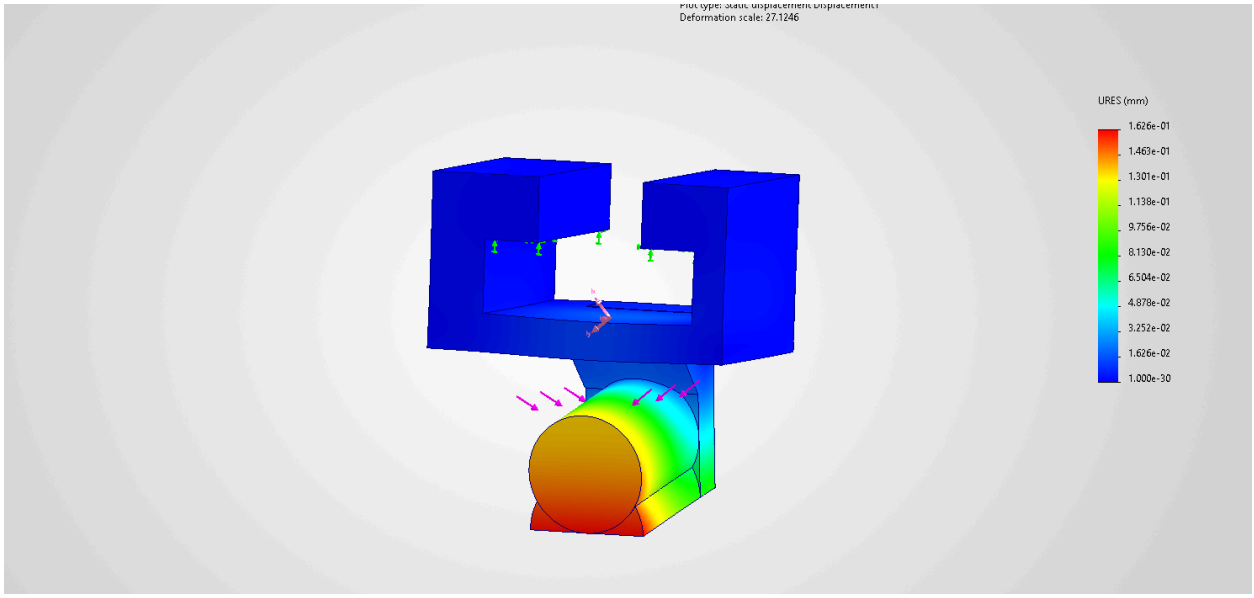


FOS

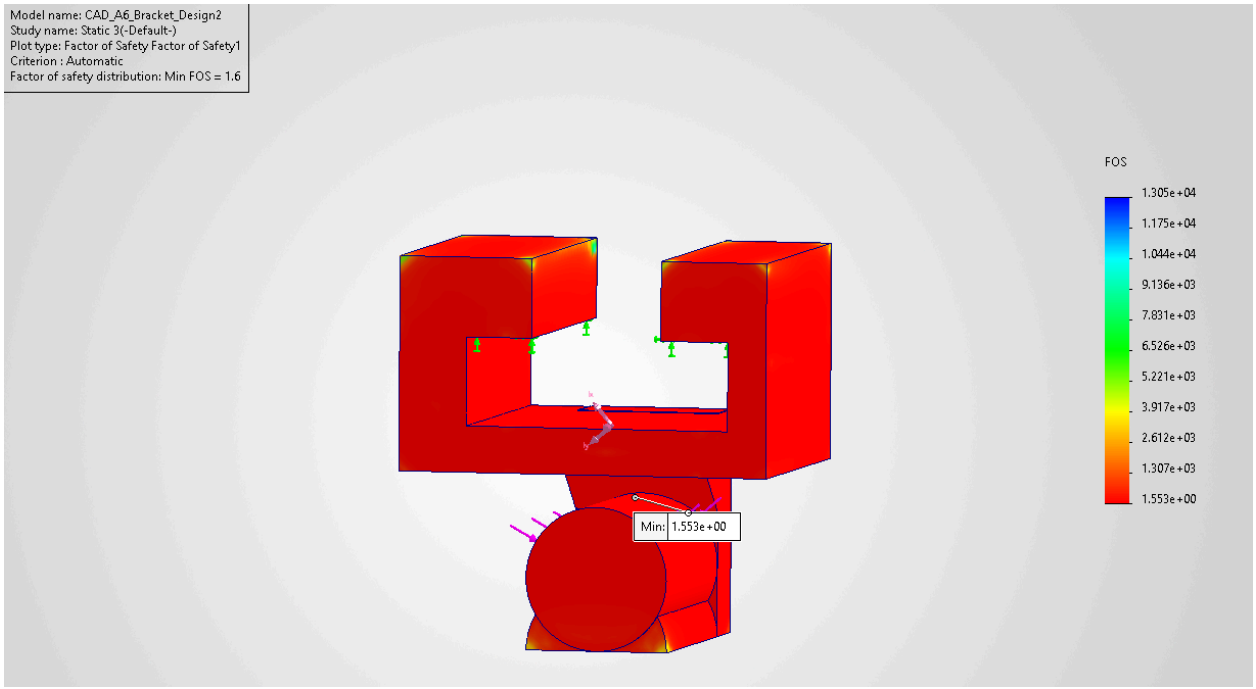
Fine mesh:



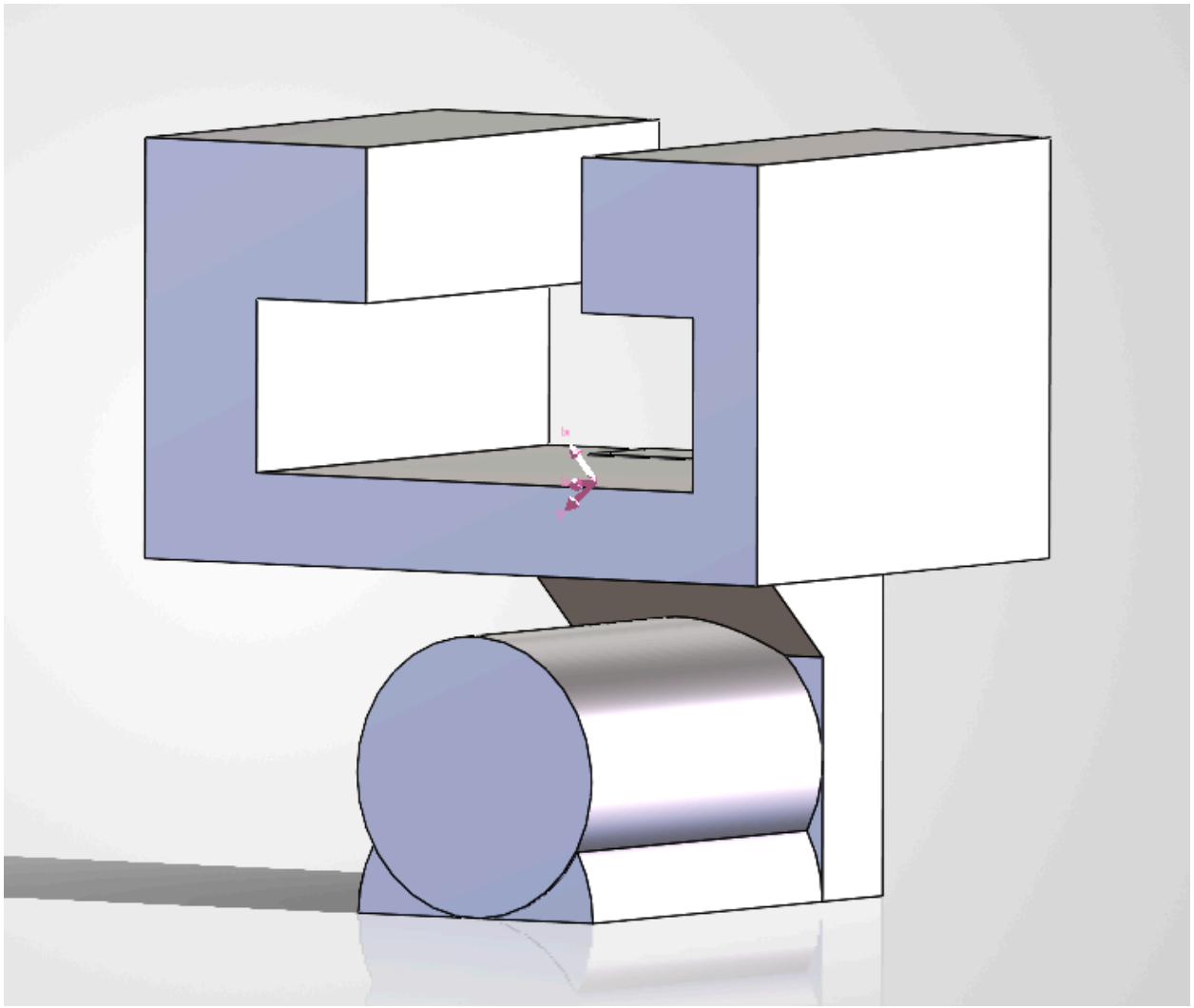
Stress



Deflection



FOS



Final Design